



B51EL: Compressible Fluids, Major Assignment

Group ED18

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Date: 23/04/2026

Abstract

With recent legislative changes and a renewed interest in supersonic flight, the desire to fly passengers around the world at supersonic speeds is leading engineers to develop new advanced designs to achieve this. This report outlines the proposed design for an engine of a supersonic transport aircraft to operate over a range of conditions using both SCRAM and RAM engines.

It outlines the theoretical basis for the design for the key components of the engine; the inlet, diffuser, combustion chamber and Converging-Diverging Nozzle. It also includes some experimental validation, however issues with the experiment meant that full validation could not be achieved. It also outlines the steps that would need to be taken to convert the aircraft to run on sustainable aviation fuels.

Table of Contents

1	Introduction.....	6
2	Experimental Data Gathering	8
2.1	Introduction.....	8
2.2	Theory	8
2.3	Experimental method	9
2.3.1	Equipment.....	9
2.3.2	Methodology.....	10
2.4	Results	10
2.4.1	Mach Number	11
2.4.2	Pressure	13
2.4.3	Mass Flow Rate	14
2.5	Discussion	14
2.6	Experimental Conclusions	16
3	Design.....	17
3.1	Engine Inlet.....	17
3.2	Initial Diffuser Compression	23
3.2.1.	Design Assumptions and Method	23
3.2.2.	Stagnation Conditions	24
3.2.3.	Static Flow Properties.....	24
3.2.4.	Diffuser Area Ratio	24

3.2.5.	Continuity Verification	25
3.2.6.	Discussion.....	25
3.2.7.	Limitations	26
3.3	Combustion Chamber	26
3.3.	Converging-Diverging Nozzle	35
3.3.1.	Freestream conditions.....	35
3.3.2.	Minimum Acceleration Force	35
3.3.3.	Design Exit Mach Number at Cruise	36
3.3.4.	Nozzle Area Ratio	36
3.3.5.	Exit Temperature	36
3.3.6.	Exit Velocity	36
3.3.7.	Required Mass Flow Rate.....	37
3.3.8.	Throat Area from Choking	40
3.3.9.	Circular Diameters	40
3.3.10.	Validation	40
3.3.11.	Engine Flow Property Profile and Expansion Status	43
3.3.12.	Minimum Altitude to Avoid Internal Discontinuities.....	46
3.3.13.	Results and Discussion	47
3.3.14.	Model Limitations	48
3.4.	Conversion to Sustainable Fuel.....	49
4.	Comparison to Experimental Data.....	50
5.	Conclusion	50
6.	Appendices	51
	Appendix 1 – Contribution Table	51
	Appendix 2 - Raw Experimental Data.....	51
	Appendix 3 - CD Nozzle Sample Calculations	51
	Appendix 4 - Diffuser.....	54
	Appendix 5 - SAF Calculations	57
7.	References	58

Table of Figures

Figure 1: Rayleigh Flow T-S Graph [5].	7
Figure 2: Equipment used in the experiment.	9
Figure 3: Nozzle diameters and positional data [7].	10
Figure 4: Experimental Mach Number through the nozzle from the inlet (at 0mm) to the outlet.	11
Figure 5: Theoretical Mach Number through the nozzle from the inlet (at 0mm) to the outlet.	11
Figure 6: Inlet flow velocity vs Exit pressure.	12
Figure 7: Measured pressures across the Nozzle, from the inlet at 0 to the outlet.	13
Figure 8: Theoretical pressures across the Nozzle, from the inlet at 0 to the outlet.	13
Figure 9: Experimental Mass Flow Rate through the Nozzle.	14
Figure 10: Diagram of flow behaviour in Ramjet configuration. Flow path in blue, oblique shock in single red lines, normal in double red lines, and geometry in black. Delta denoted the wedge angle, Beta the angle of the first oblique shock.	18
Figure 11: Diagram showing shock chain configuration for Scramjet. Delta remains the same as for the Ramjet. The column is shown extended to account for the change in regimes.	19
Figure 12: Achievable pressures for both Scramjet (a) and Ramjet (b).	21
Figure 13: Overlaid achievable pressures.	21
Figure 14: Expected achievable pressure curves under reasonable requirements.	22
Figure 15: RAM Mass Flow Rate Across Outlet Mach Range	32
Figure 16: SCRAM Mass Flow Rate Across Outlet Mach Range	33
Figure 15: Mass Flow Rate Sensitivity to Nozzle Inlet Conditions- Scramjet Mode.	39
Figure 16: Mass Flow Rate Sensitivity to Nozzle Inlet Conditions- Ramjet Mode.	39
Figure 17: Engine flow property profile - station by station.	44
Figure 18: Exit velocity and expansion status comparison.	45

Table of Nomenclature

Symbol	Units	Definition
A_e	m^2	Nozzle exit area
A_t	m^2	Nozzle throat area
D_e	m	Nozzle exit diameter
D_t	m	Nozzle throat diameter
F_{acc}	N	Minimum acceleration force required
L_c	m	Critical length
M_0	—	Freestream Mach number
M_e	—	Exit Mach number
M_t	—	Throat Mach number
$T_{0,5}$	K	Nozzle-inlet stagnation temperature at station 5
T_0	K	Stagnation temperature
T_a	K	Ambient temperature
T_e	K	Nozzle exit static temperature
$\dot{m}$	kg/s	Mass flow Rate
$p_{0,5}$	Pa	Nozzle-inlet stagnation pressure at station 5
p_a	Pa	Ambient pressure
p_e	Pa	Nozzle exit static pressure
p_o	Pa	Stagnation pressure
u_0	m/s	Freestream velocity
u_e	m/s	Nozzle exit velocity
v_0	m^3/kg	Specific Stagnation Volume
ΔV	m/s	Change in aircraft velocity
A	m^2	Area
A/A^*	—	Area ratio relative to sonic throat area
D	m	Diameter
F	N	Thrust
Ma	—	Mach Number
T	K	Static temperature
V	m/s	Aircraft velocity
g	m/s^2	Gravitational acceleration
p	Pa	Pressure
t	s	Acceleration time
u	m/s	Flow velocity
β	$^\circ$	Oblique shock wave angle
γ	—	Adiabatic Index
γ	—	Specific heat ratio combustion mixture
δ	$^\circ$	Wedge angle
ρ	kg/m^3	Density

1 Introduction

The use of supersonic engines in commercial flights has had an extensive and often controversial history. The many challenges and obstacles which surround the field have led to the current absence of flights which operate at a speed of over Mach 1, with the U.S Federal Aviation Administration driving the global policy of an over-land ban [1] on commercial flight shortly after its introduction. However, in recent years there has been a resurgence in the interest of commercial supersonic flight; advancements in materials, propulsion efficiency and aerodynamics paired with the recent repeal of the ban of civilian supersonic flight over land in the United States [2] means that projects such as the Boom Overture [3] are pushing the field back into the realm of practicality once again.

There are three main types of supersonic engines: turbojet, ramjet, and scramjet. The turbojet uses a compressor to feed compressed air into the combustion chamber, as opposed to the ramjet and scramjet which use their inlet geometry to create shockwaves and compress air naturally when travelling at speed. Ram/scramjets cannot provide enough static thrust to take off themselves. Design of the inlet is critical as the shocks in the system increase the static pressure and temperature of the incoming air while reducing the Mach number. This allows for efficient combustion and improved pressure recovery as losses from shock waves are reduced before the combustion chamber. Fanno flow principles must also be considered, as large duct lengths and rough surfaces can lead to significant performance losses, as friction in the ducts reduces stagnation pressure and pushes Mach number to 1.

With the use of CFD, researchers are attempting to minimise disruptive sonic booms, predicting their placements and using this data to refine the aircraft's body, angling the wave away from the ground [4]. Maximising engine fuel efficiency and thrust across all speed regimes while minimising operational costs also poses a hurdle to the reintroduction of supersonic civil flight. Therefore, Rayleigh flow analysis must be performed to optimise the material and flow conditions of the combustion chamber and avoid thermal choking which will alter the Mach number. The Rayleigh Flow T-S cycle is shown in Figure 1.

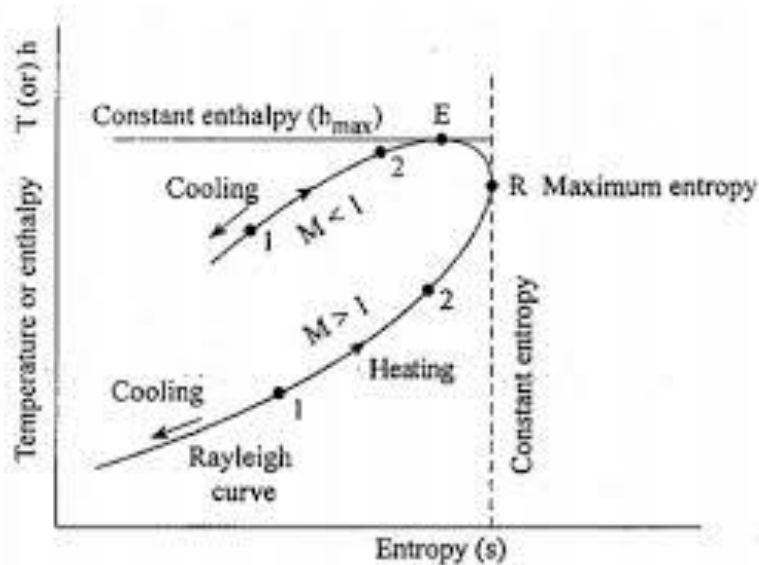


Figure 1: Rayleigh Flow T-S Graph [5].

Also, correct nozzle expansion is critical in converting the thermal energy into kinetic and ensuring optimal exhaust velocity and thrust generation, and so must be considered.

The most prominent supersonic commercial aircraft model of all time is that of Sud Aviation and British Aircraft Corporation's (currently Airbus and BAE systems, respectively) Concorde. The aircraft utilised four Rolls-Royce/Snecma Olympus 593 Mk 610 afterburning turbojet engines with variable geometry intakes designed to generate multiple oblique shockwaves for compression at a cruise speed of up to Mach 2.04 and a maximum thrust generation of 169.2 kN per engine [6]. This slowed incoming air at this speed to subsonic before the compressor and maximised pressure recovery at all Mach regimes. Regardless of its technical success, high operating costs paired with the Air France Flight 4590 tragedy in 2000 forced the Concorde into retirement.

The analysis of compressible flow is integral for the development of supersonic engines in all fields. The manipulation of shockwaves, thermal choking and limitations of materials, and viscous flow through pipes and nozzle expansion all determine the performance of supersonic aircraft. With the key constraints of the great operating costs and decibel levels of the sonic booms produced experiencing recent advancements, supersonic commercial flights may soon be feasible again.

This report outlines the theoretical and experimental basis for a proposed RAM/SCRAM Supersonic Transport Jet and gives some guidance on the proposed conversion to sustainable fuels to help alleviate the concerns around the emissions from flights in general and supersonic flight in particular.

2 Experimental Data Gathering

2.1 Introduction

Understanding the flow through a Convergent - Divergent nozzle is critical to understanding the behaviour of practical supersonic systems. While lots can be learned from theoretical and computational methods, the complexity of the flow patterns can only fully be understood with the aid of experiments. To help verify and validate the underlying theory used in the design process below small-scale experiments using air were conducted to demonstrate the flow conditions in terms of pressure and mass flow rate for choked, subsonic and supersonic flow.

2.2 Theory

Flow through a Convergent-Divergent Nozzle is well understood and as such can be analysed using the same formulas utilised later in the report to design the Nozzle for the jet engine. To convert the experimental pressure measured in the experiment to a Mach Number Equation 1 is used. γ is assumed to be 1.4 as the fluid used is air.

$$\frac{p_0}{p} = \left(1 + \frac{\gamma - 1}{2} Ma^2\right)^{\frac{\gamma}{\gamma - 1}} \quad (1)$$

The stagnation pressure must also be calculated using the conditions at the inlet where both Ma and p are known, as shown in Equation 2 below. This stagnation pressure can then be used across the full nozzle as it is assumed that the process is isentropic.

$$p_0 = \left(1 + \frac{\gamma - 1}{2} Ma^2\right)^{\frac{\gamma}{\gamma - 1}} \cdot p = \left(1 + \frac{1.4 - 1}{2} 0.022442^2\right)^{\frac{1.4}{1.4 - 1}} = 101335.7163 Pa \quad (2)$$

To calculate the theoretical Mach Number, area ratio data from standardised tables of various fluid properties for a range of Ma were used, in particular the area ratio between the area of interest and the preceding area.

The theoretical pressure was calculated using Equation 1 again, and the calculated Mach numbers.

The remaining properties were then extracted from both sets of results using similar equations, the mass flow rate was calculated using Equation 4, with the required specific stagnation mass calculated using Equation 5. r_p is defined by Equation 6.

$$\dot{m} = \left[\frac{2\gamma p_0}{(\gamma - 1)v_0} \right]^{\frac{1}{2}} (r_p)^{\frac{1}{\gamma}} \left(1 - (r_p)^{\frac{\gamma-1}{\gamma}} \right)^{\frac{1}{2}} A \quad (4)$$

$$v_0 = \frac{1}{\rho_0} = \frac{1}{\left(1 + \frac{\gamma-1}{2} Ma^2 \right)^{\frac{1}{\gamma-1}}} \quad (5)$$

$$r_p = \frac{p}{p_0} \quad (6)$$

2.3 Experimental method

2.3.1 Equipment

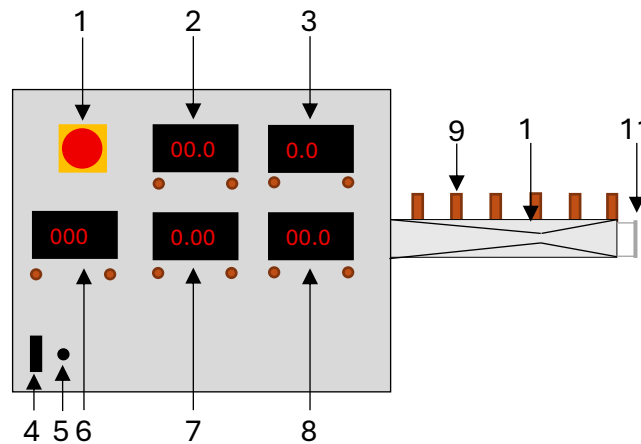


Figure 2: Equipment used in the experiment

The experiment was conducted using an M 230 Flow of compressible fluids from GUNT depicted in Figure 2. Air was sucked into the machine through a guarded inlet (11), this air then passed through the convergent – divergent nozzle (10). The suction force was provided by a compressor that could have its rpm monitored and adjusted through the control panel (5). To monitor the pressure there was several holes with a quick disconnect fitting on the top along the length of the nozzle (9), these outlets could then be connected to the control panel to give an output of pressure in mbar or bar, or the flow velocity in m/s (2,3 and 6,7,8). The quick disconnect fittings (7) meant that the pipes could be moved between the different monitoring positions when the experiment was running allowing the pressure along the length of the nozzle at different flow rates to be monitored. For safety the compressor could be isolated using the e-stop (1) and the on off switch (4).

2.3.2 Methodology

Prior to starting the experiment the correct PPE was worn, namely ear protection. The following method was then followed.

1. The pipes were connected to measure the pressure and the inlet velocity.
2. The compressor was turned on using the isolation switch (1) and then the on off switch (4)
3. The compressor was then set to the initial to the initial rpm of 4000 using the rpm adjustor (5)
4. The pressure was then measured at all the positions across the convergent-divergent nozzle (10) by connecting the pipe to the various quick connect fittings (9) to the ports on the control panel.
5. The RPM was then increased by 2000rpm and step 4 was repeated.
6. Step 5 was repeated until the max rpm of 23500 was reached.
7. The pressure and inlet velocity were then recorded at each stage.
8. Once the experiment was completed the propellor was turned off using the on off switch (4) followed by the isolator (1)

2.4 Results

The graphs below show the key results; the raw measured pressures can be found in Appendix 2. A diagram of the nozzle and its dimensions is shown in Figure 3.

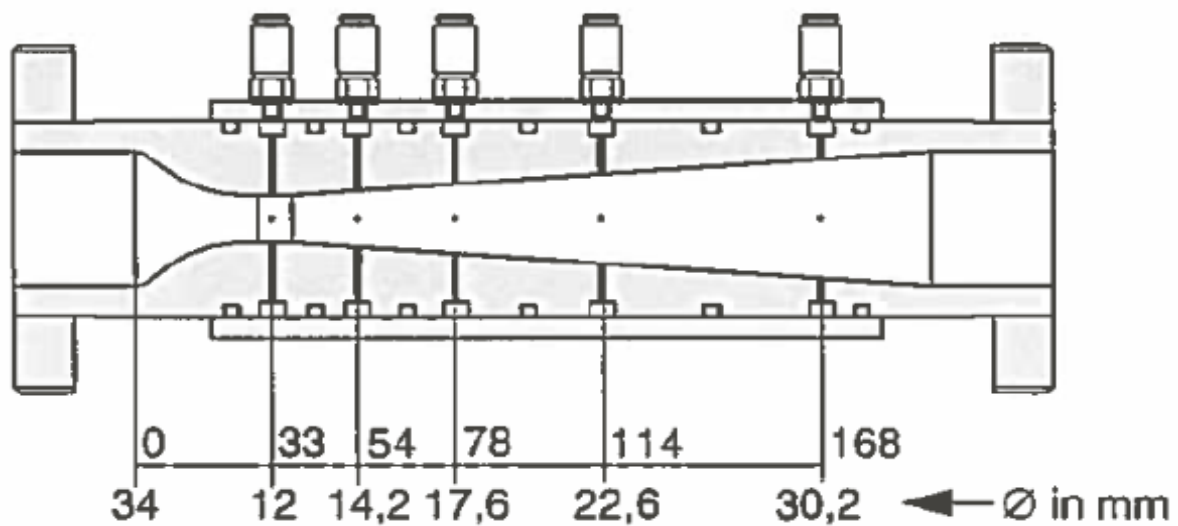


Figure 3: Nozzle diameters and positional data [7]

2.4.1 Mach Number

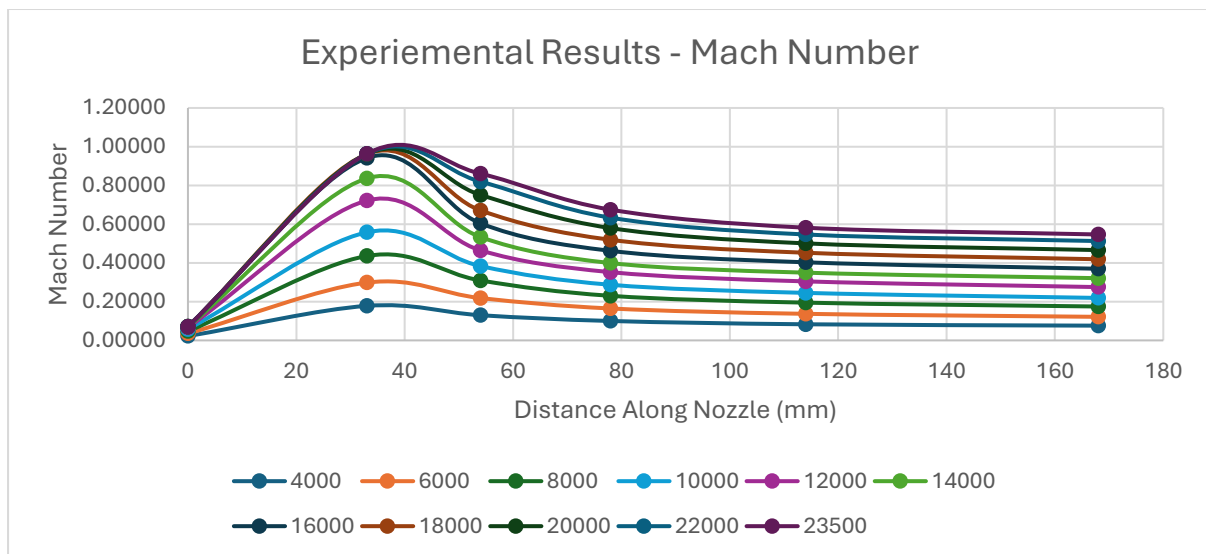


Figure 4: Experimental Mach Number through the nozzle from the inlet (at 0mm) to the outlet.

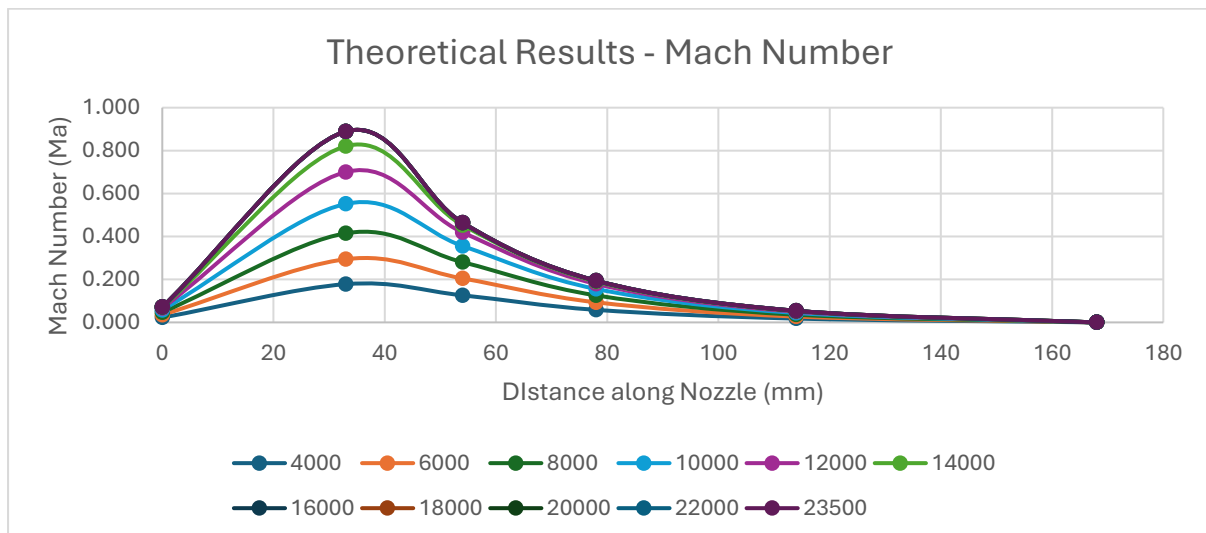


Figure 5: Theoretical Mach Number through the nozzle from the inlet (at 0mm) to the outlet.

As can be seen from both the experimental (Figure 4) and theoretical results (Figure 5), both remain subsonic and reach their peak at the throat as would be expected for a subsonic flow in a convergent divergent nozzle. The most marked difference comes from the marked difference as the flow goes through the outlet. The decrease for the theoretical Mach number is much faster across all flow speeds, decreasing to a negligible speed by the time it reaches the outlet. In contrast, the speed for the experiment remains high (at 0.6 at the outlet).

This is further seen in the results in Table 1, where the percentage widens significantly as the flow move towards the tail, it is also worth noting that while choked flow was observed at the inlet (at an inlet Mach of 0.072) the experimental results still shown an increase in Mach number, differing from the theory that does predict the choked flow preventing any further increase in flow velocity.

Table 1: Theoretical and experimental data for the Mach Number results.

Distance (mm)	Theoretical Mach (-)					Experimental Mach (-)					Percentage Change (%)				
	33	54	78	114	168	33	54	78	114	168	33	54	78	114	168
0.022	0.18	0.13	0.06	0.02	0.00	0.18	0.13	0.10	0.08	0.08	0%	3%	42%	79%	100%
0.036	0.29	0.20	0.09	0.03	0.00	0.30	0.22	0.16	0.14	0.12	1%	6%	43%	80%	100%
0.047	0.41	0.28	0.13	0.04	0.00	0.44	0.31	0.23	0.19	0.17	5%	9%	45%	82%	100%
0.058	0.55	0.36	0.16	0.04	0.00	0.56	0.38	0.29	0.25	0.22	1%	7%	46%	82%	100%
0.066	0.70	0.42	0.18	0.05	0.00	0.72	0.46	0.35	0.30	0.28	3%	10%	49%	84%	100%
0.070	0.82	0.45	0.19	0.05	0.00	0.84	0.53	0.40	0.35	0.32	2%	15%	52%	85%	100%
0.072	0.89	0.46	0.19	0.05	0.00	0.94	0.60	0.46	0.40	0.37	6%	23%	58%	87%	100%
0.072	0.89	0.46	0.19	0.05	0.00	0.96	0.67	0.52	0.45	0.42	8%	31%	63%	88%	100%
0.072	0.89	0.46	0.19	0.05	0.00	0.96	0.75	0.58	0.50	0.47	8%	38%	66%	89%	100%
0.072	0.89	0.46	0.19	0.05	0.00	0.96	0.82	0.63	0.55	0.51	8%	43%	69%	90%	100%
0.072	0.89	0.46	0.19	0.05	0.00	0.96	0.86	0.67	0.58	0.55	8%	46%	71%	91%	100%

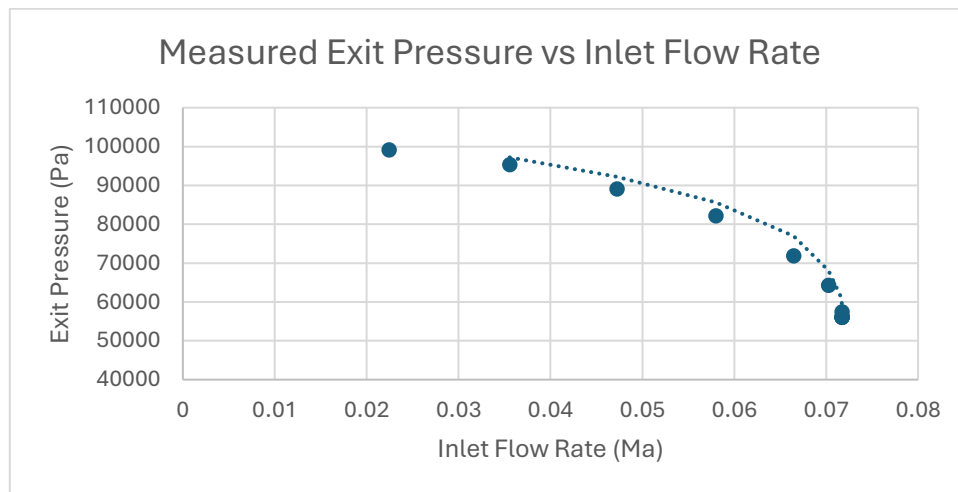


Figure 6: Inlet flow velocity vs Exit pressure

Figure 6 clearly shows the expected trend that as the inlet Mach number and therefore the Mach number through the system increases the exit pressure decreases, as would be expected. As the exit inlet Mach starts to become choked and the rate of increase slows then the outlet pressure shows a similar trend, dropping at a steadily decreasing rate until remaining near equal when the flow does become choked.

2.4.2 Pressure

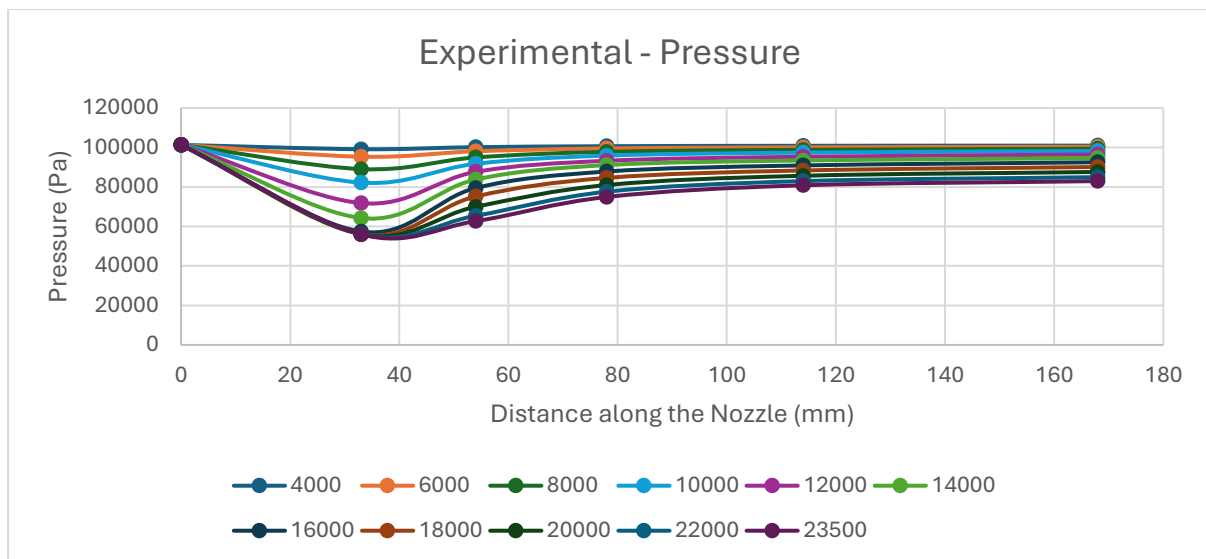


Figure 7: Measured pressures across the Nozzle, from the inlet at 0 to the outlet.

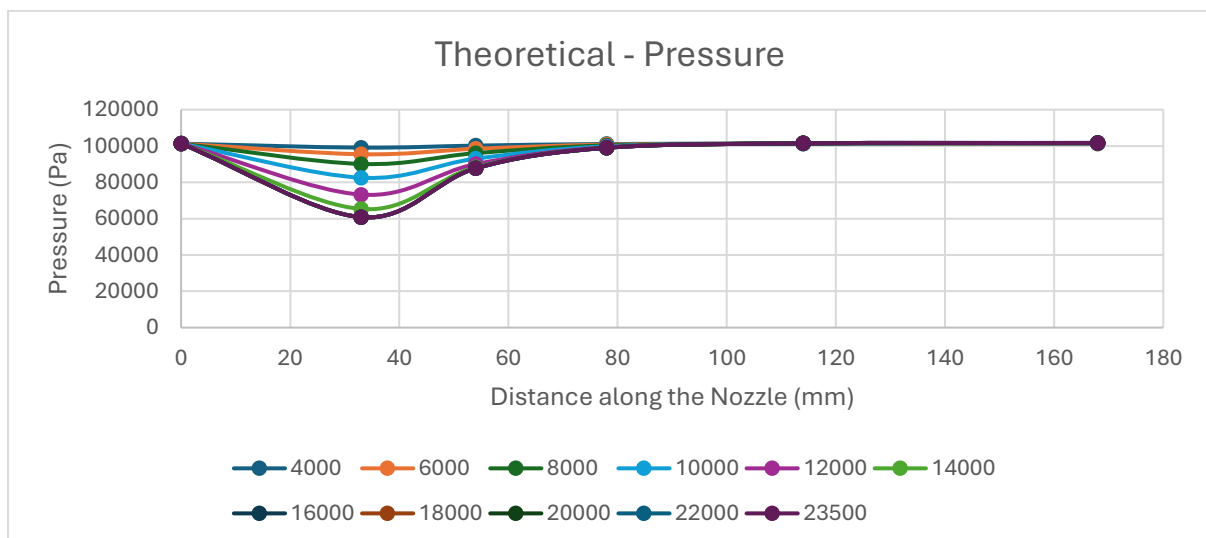


Figure 8: Theoretical pressures across the Nozzle, from the inlet at 0 to the outlet.

The pressure graphs (Figure 7 and Figure 8) show a similar trend to that of the Mach results with again a much faster increase to towards atmospheric pressure than the experimental results show, the minimum pressure reaches it's minimum at the throat, aligning with the Mach Number results. Similarly to the Mach numbers the same pattern of the trends matching much closer before the throat is continuing. Again, the minimum values also show a marked difference with an increasing difference as the flow approaches the outlet.

2.4.3 Mass Flow Rate

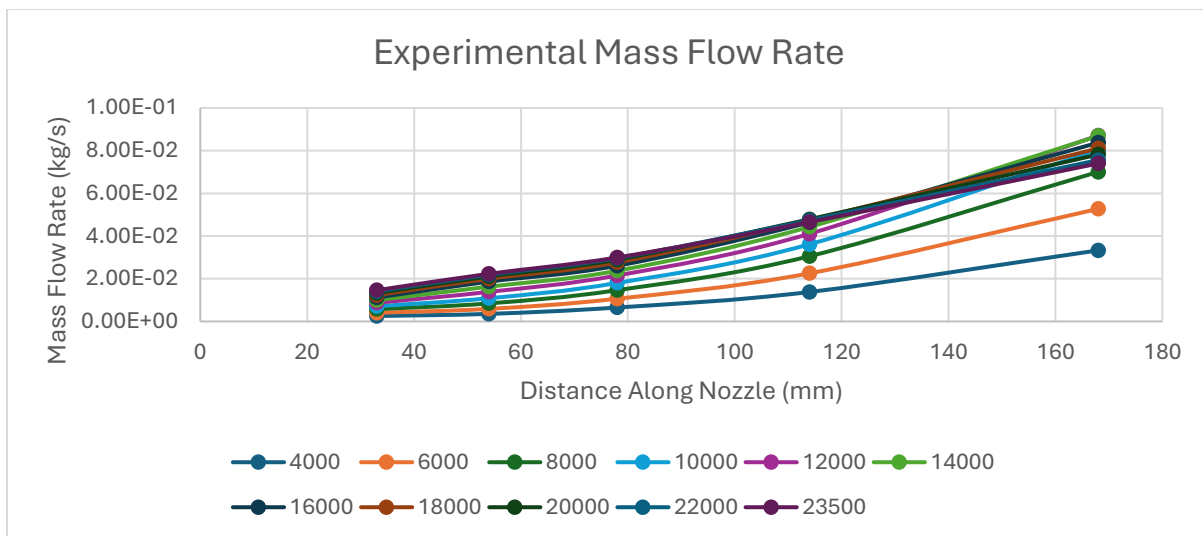


Figure 9: Experimental Mass Flow Rate through the Nozzle.

The most marked difference between theoretical and experimental behaviour is seen in the mass flow rate. It would be expected that given there is no additional inputs to the system in terms of air volume that the mass flow rate should remain constant. However it can be seen in Figure 9 that the mass flow rate increases significantly across the nozzle, for each of the fan speed settings.

2.5 Discussion

It can be seen between the experimental and theoretical trend a clear pattern emerges. While the trend exhibited is broadly similar of an increase in velocity as the nozzle converges and then as it diverges a subsequent decrease in velocity along with an associated drop and then rise in pressure. The rate of decrease when it passes through the throat is far faster in the theoretical results than the experimental results, increasing to a 100% difference when the theoretical model predicts a negligible velocity at the tail and the experimental model still shows a relatively high Mach number

One possible explanation for this divergence from theory is that the compressor was placed at the outlet of the nozzle, meaning that the flow would still be acted on by the compressor as it exits the throat.

Overall the Mach number predicted by theory is far higher, leading to the predicted choked flow, while it is not possible to conclude on the exact reason for this the theoretical model does not account for friction along the walls of the nozzle which would remove energy from the flow and thus decrease the flow velocity. It is possible to adjust the theoretical model to account for this using Fanno flow theory but as the factor of friction for the nozzle is unknown it would not be possible to in this case.

The aim of the experiment was to analyse the subsonic, choked and supersonic flow through a Convergent-Divergent nozzle, it is worth noting that while the inlet velocity and the theoretical results seen in Table 1 suggest that choked flow, should be achieved, the calculated experimental Mach numbers and pressures suggest this was not the case.

At choked flow the mass flow rate will not increase any further when the flow through the throat becomes sonic ($Ma=1$). While the theoretical model does show that the Mach number at the throat does reach close to Mach 1, and the flow subsequently being measured as choked (in reality likely close to choked), the experimental results show a clear trend of an increase in Mach number. Further correlation for this is shown in Figure 6 that clearly demonstrates that the expected relationship between exit pressure and inlet Mach is followed.

Turning to supersonic flow, the flow in both the theoretical and experimental results did not reach supersonic speeds, this is likely due to power limitations with the compressor meaning that there is not enough power to accelerate the flow to the speeds needed to reach supersonic flow. The theoretical model as seen later in the design section would predict a further increase in Mach number as the flow moves past the throat and along the outlet. Given that the results do not show this trend at all, it provides further evidence that the flow did not reach the supersonic speeds needed to validate this aspect of flow through the nozzle further.

One final clear indication that there is a significant fault with the experimental apparatus is the mass flow rate results. Theory would predict that so long as there is no additional mass added to the system then the flow rate through the nozzle would remain constant, however it can be seen from Figure 9 that this is clearly not the case. This suggests that there is more mass inlets to the nozzle than just that at the inlet. Given that the experimental apparatus is connected to the measuring equipment by quick disconnect ports there is a possibility that these are the source of the leak, however without monitoring the flow at each location it is not possible to know.

Due to time restrictions this experiment was not able to be repeated, which would have confirmed whether the trends and issues observed were systematic or only in this one experiment. Further value to confirm the correlation in trends between the theoretical and experimental methodologies could also be found by repeating the experiment on a nozzle with a different geometry to see if the trends are repeated.

2.6 Experimental Conclusions

Overall, the significant issues encountered in the experiment in particular the indication of a leak in the system and the arrangement of the compressor prevented the experimental data from providing as much insight and verification as was hoped to the design stage of the project. Nevertheless, the study did validate the theoretical model in the subsonic regime showing a clear trend of increasing Mach number towards the throat and a subsequent deceleration as the flow travels to the outlet.

As the flow failed to reach the choked or supersonic regimes, it is impossible to validate them further limiting its effectiveness in evaluating the design methodology.

3 Design

Equations used in the design process are taken from the course notes [8] and formula sheet [9] unless otherwise stated.

3.1 Engine Inlet

Table 2: Required flow conditions for the inlet stage

Configuration	Scramjet	Ramjet
Mach In	3.02	2.62
Mach Out	1.52	0.76
Pressure Out (kPa)	173	324
Altitude (ft -- m)	65500 -- 19964	40500 -- 12344

For both Scramjet and Ramjet configurations, the fluid must be slowed down for effective mixing and combustion in the later stages of the engine. The inlet aims to begin this process by passing the incoming flow through several shock waves to greatly reduce the speed and increase the pressure of the incoming fluid. When fluid passes through a shock wave, the flow undergoes a change from supersonic to subsonic conditions, with a sudden increase in pressure. Normal shock waves are the easiest configuration to consider this in. Normal shock waves are caused by a sufficiently high back pressure such that a sudden pressure step is required out with regular flow conditions.

Oblique shock waves differ slightly from normal shock waves, in terms of how they are caused. When supersonic flow encounters an angled surface, the fluid at the interface is slowed by the obstruction and heavily compressed as it changes direction into other fluid. This results in compression waves as the fluid turns, also known as oblique shock.

As mentioned, when flow passes through a shock wave it goes from supersonic to subsonic. However, oblique shock waves do not form perpendicular to the flow – as such, only the velocity component of the flow perpendicular to the shock will be reduced to subsonic – the tangential component may remain supersonic, resulting in the overall velocity of the flow remaining supersonic. Therefore, by controlling the angle at which the oblique shock waves form by controlling the wedge angle, the resultant velocity decrease and related pressure increase are also controlled.

Once the flow has passed the wedge and thus the oblique shocks, the flow will enter the inside of the inlet. This inside of the inlet is simplified 2-dimensions and as such is treated as two straight pipes, visualised in Figure 10. The flow through these pipes is governed by Fanno flow, which describes how compressible fluid moves through straight pipes. If upon entry to the Fanno flow section the flow is subsonic, it will increase to approach choking at Mach 1. Alternatively, if the flow is supersonic it will decrease to choking at Mach 1.

This section is crucial for controlling the differences between Scramjet and Ramjet. Because Ramjet is expected to exit the inlet at subsonic speeds, it will be important to pass it through a terminal normal shock wave – while avoiding doing the same to the Scramjet. It is in this section that variable geometry will be implemented. As demonstrated by the SR-71 Blackbird [10], by extending and retracting the central column on which the wedge is mounted, the length of the Fanno flow section can be controlled – allowing for correct adjustment of the Mach number before the exit, and avoidance of normal shock being reached in Scramjet mode.

In this design, the back pressure for both Scramjet and Ramjet modes is provided by the subsequent sections. Therefore, it is possible to fine tune the designed wedge angle and the adjustable Fanno flow section to cause the desired Mach number decrease, while working backwards to require an inlet pressure that would be expected of the operating altitude.

The shock chain used is detailed in Figure 10 and Figure 11. For both configurations, there are two shocks – one of which is caused by the flow deflecting, the other by the flow correcting – followed by a Fanno flow segment. For ramjet, this is then ended at a terminal normal shock. For Scramjet, this is avoided.

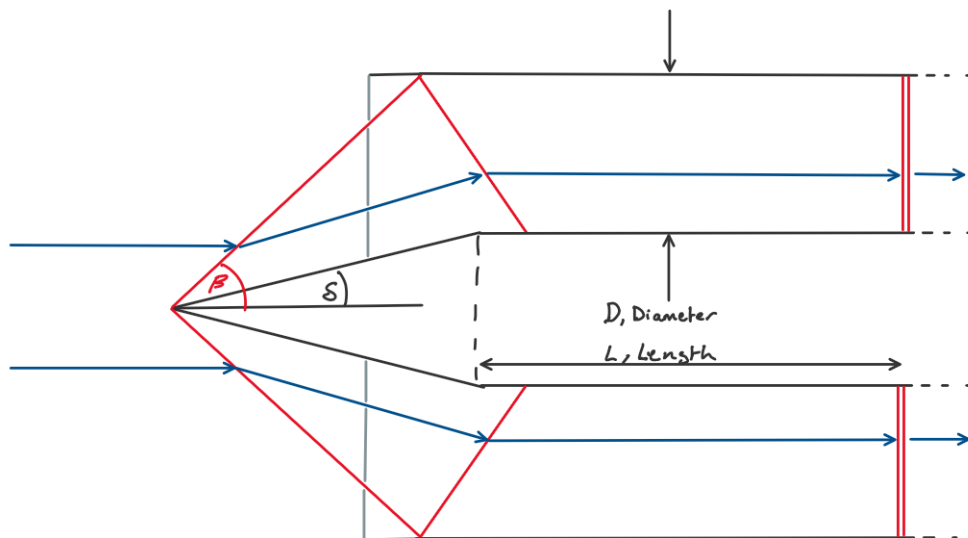


Figure 10: Diagram of flow behaviour in Ramjet configuration. Flow path in blue, oblique shock in single red lines, normal in double red lines, and geometry in black. Delta denoted the wedge angle, Beta the angle of the first oblique shock.

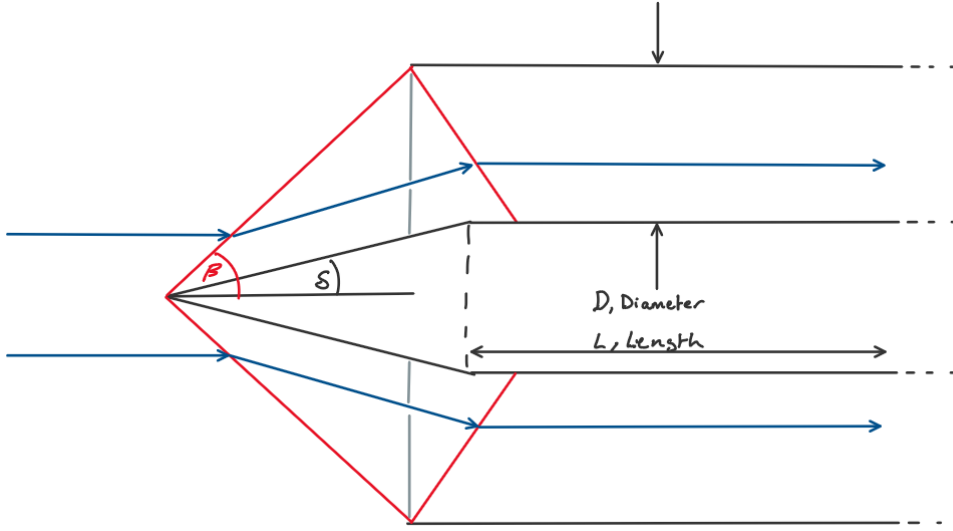


Figure 11: Diagram showing shock chain configuration for Scramjet. Delta remains the same as for the Ramjet. The column is shown extended to account for the change in regimes.

The total pressure at the entrance to the inlet is given by considering both the static pressure at the altitude and the dynamic pressure caused by the movement of the jet. These can be calculated by using techniques from NASA for atmospheric conditions in the lower stratosphere. [11, 12] seen in Equations 7 to 11.

$$T_{atm} = -56.46 \text{ }^{\circ}\text{C}$$

$$P_{Static,atm} = 22.65 e^{(1.73 - 0.000157 \times \text{Altitude})} \quad (7)$$

$$\rho_{atm} = \frac{P_{Static,atm}}{(0.2869 (T_{atm} + 273.1))} \quad (8)$$

$$u_{jet} = Ma_{jet} \sqrt{\gamma R (T_{atm} + 273.1)} \quad (9)$$

$$q = \frac{\rho u^2}{2} \quad (10)$$

$$P_{Total} = P_{Static,atm} + q \quad (11)$$

Where ρ_{atm} is the density of the atmosphere at this altitude, u_{jet} is the velocity of the jet in m/s, Ma_{jet} is the Mach number of the jet, γ is the adiabatic index of air, R is specific gas constant and q is the dynamic pressure. This gives 41 and 107 kPa for Scramjet and Ramjet respectively.

The flow properties through oblique shock waves can be calculated using equations 12-16 and table values.

$$Ma_{1normal} = Ma_1 \sin \beta \quad (12)$$

$$Ma_{tangential} = Ma_1 \cos \beta \sqrt{\frac{\rho_2/\rho_1}{p_2/p_1}} \quad (13)$$

$$Ma_2 = \sqrt{Ma_{2normal}^2 + Ma_{tangential}^2} \quad (14)$$

Where the normal component is calculated as with normal shock calculations.

$$\frac{\rho_2}{\rho_1} = \frac{(\gamma + 1)Ma_1^2}{2 + (\gamma - 1)Ma_1^2} \quad (15)$$

$$\frac{p_2}{p_1} = \frac{2\gamma Ma_1^2 \sin^2 \beta - (\gamma - 1)}{\gamma + 1} \quad (16)$$

Flow properties through normal shock waves can be calculated using equations 17 and 18 and table values.

$$Ma_2^2 = \frac{1 + \frac{\gamma - 1}{2} Ma_1^2}{\gamma Ma_1^2 - \frac{\gamma - 1}{2}} \quad (17)$$

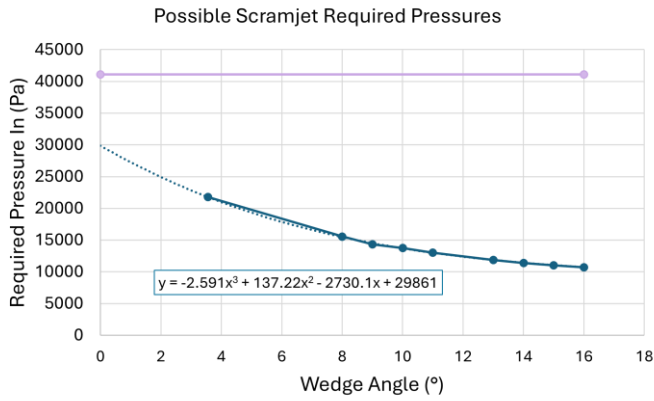
$$\frac{p_2}{p_1} = \frac{1 + \gamma Ma_1^2}{1 + \gamma Ma_2^2} \quad (18)$$

Flow properties through the Fanno flow section can be calculated with equations 19 and 20 and table values.

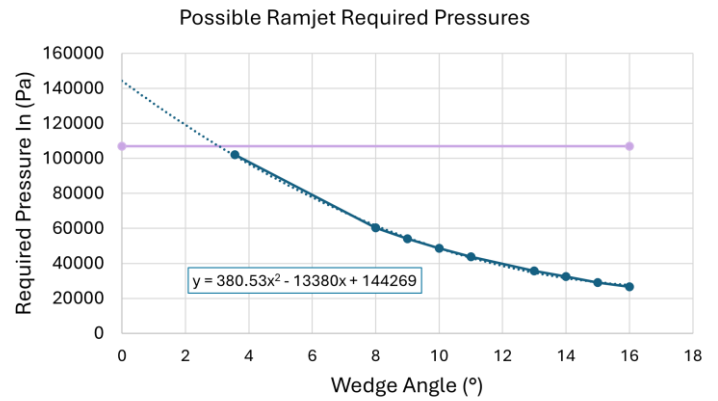
$$\frac{p_2}{p_1} = \frac{Ma_1}{Ma_2} \left(\frac{1 + \frac{\gamma - 1}{2} Ma_1^2}{1 + \frac{\gamma - 1}{2} Ma_2^2} \right)^{\frac{1}{2}} \quad (19)$$

$$4f \frac{L_c}{D} = -\frac{1}{\gamma} \left(1 - \frac{1}{Ma^2} \right) + \frac{\gamma + 1}{2\gamma} \ln \left[\frac{\frac{\gamma + 1}{2} Ma^2}{1 + \frac{\gamma - 1}{2} Ma^2} \right] \quad (20)$$

To meet the required pressures, design iteration must be performed. By ensuring the Mach number is always reached post-shocks by adjusting the length of the Fanno flow segment to compensate for not reaching with oblique shocks alone, the curves shown in Figure 12 are developed for Ramjet and Scramjet. This clearly shows where each configuration will match both the existing entrance and required exit values.



a)



b)

Figure 12: Achievable pressures for both Scramjet (a) and Ramjet (b).

Ramjet meets the requirements at a wedge angle δ of 3.12°, but it is not possible for Scramjet to achieve this. As the pressure at the inlet is too great for Scramjet, the behaviour would still be seen but the pressure into the diffuser would be greater than expected by a significant margin. Combining both figures results in Figure 13.

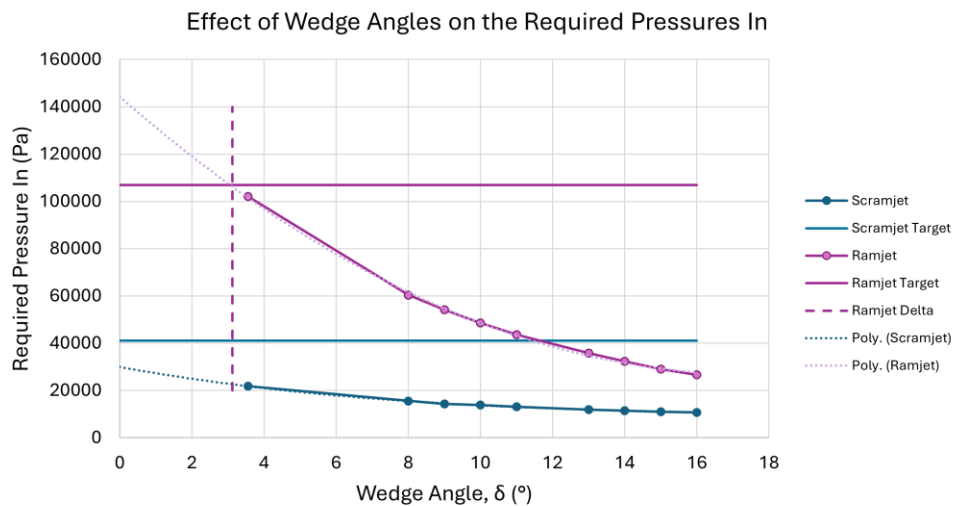


Figure 13: Overlaid achievable pressures.

Reviewing the trends in the data, with a different required pressure into the diffuser or a lower total pressure at the entrance to Scramjet, it would be expected that the ideal wedge angle for Scramjet would fall nearby that for Ramjet. In this situation, a figure similar to Figure 14 would be expected. In this case, it would be important to carefully select a value within the range to minimise the consequences to each configuration – though without variable wedge angle, it may be impossible to perfectly meet the pressure requirements at the given Mach numbers. Additionally, day-to-day atmospheric pressure and temperature differences may combine to cause fluctuations around the operating points regardless.

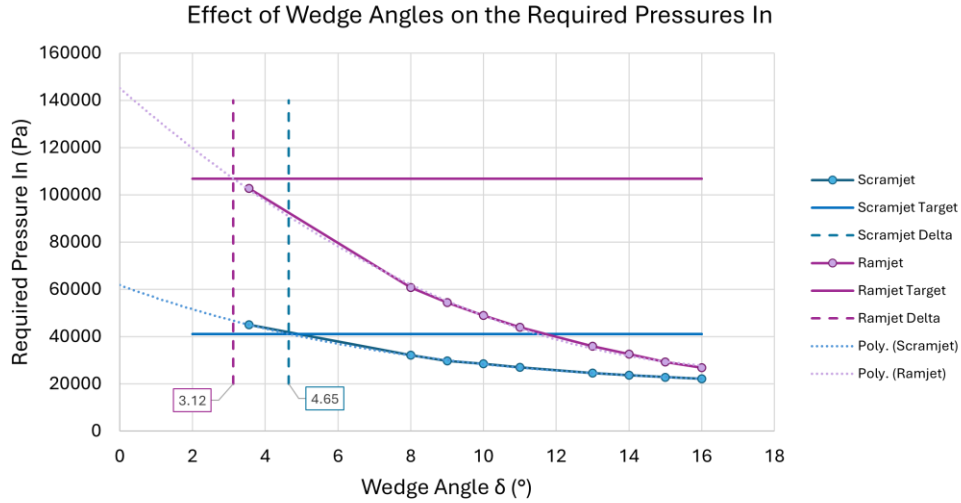


Figure 14: Expected achievable pressure curves under reasonable requirements.

Additionally, in reality there would be further limitations – the ideal wedge angle for the Ramjet configuration is 3.12°, or 6.24° degrees considering both top and bottom. This is extremely shallow and would have to be made of extremely sturdy materials to support such a geometry. Finally, the length of the following Fanno flow section has an L/D ratio of 15, requiring the engine inlet 1.5 metres in length to cause this behaviour when the outer shell has a radius of 0.1 metres more than the internal cylinder. This is not an excessively long segment, but it is longer than would typically be expected when not including the length of the wedge. Going forward, the desired not the actual values will be used for calculation in the other design sections.

3.2 Initial Diffuser Compression

After the inlet compression stage in 3.1. – the flow enters the diffuser and needs to be slowed down further before entering the combustion chamber. For ramjet mode, the diffuser reduces the Mach number from 0.76 to 0.26, while for scramjet it reduces the Mach number from 1.52 to 1.2 without the flow becoming subsonic.

3.2.1. Design Assumptions and Method

The diffuser analysis was carried out using a one-dimensional, steady, adiabatic and isentropic model with uniform flow properties at each cross-section. The working fluid was taken to have constant properties of: $\gamma = 1.3$, $R = 276 \text{ J/kg. K}$, and the duct was assumed to have a circular cross-section throughout.

The combustor inlet conditions obtained in the combustion chamber design were taken as the diffuser exit conditions, so the diffuser calculation was performed using a reverse chain approach. Because the diffuser was assumed to be adiabatic and isentropic, stagnation pressure and temperature was conserved between the diffuser inlet and outlet, therefore, $T_{0,1} = T_{0,2}$; and $P_{0,1} = P_{0,2}$.

The inlet and outlet static values were obtained from the isentropic flow correlations (Equations 21 and 22):

$$T = \frac{T_0}{\left(1 + \frac{\gamma - 1}{2} * M^2\right)} \quad (21)$$

$$P = \frac{P_0}{\left(1 + \frac{\gamma - 1}{2} * M^2\right)^{\frac{\gamma}{\gamma - 1}}} \quad (22)$$

The local speed of sound, flow velocity and density were then evaluated from the following correlations:

Speed of sound is calculated by Equation 23 :

$$a = \sqrt{\gamma RT} \quad (23)$$

The Flow velocity is calculated by Equation 24:

$$V = Ma \quad (24)$$

The density by Equation 25:

$$\rho = \frac{P}{RT} \quad (25)$$

The diffuser area change required in each operating mode was determined from the quasi-one-dimensional area-Mach correlation, further verified using continuity.

3.2.2. Stagnation Conditions

The diffuser exit stagnation conditions were taken from the combustor inlet analysis. For the scramjet and ramjet operating conditions the stagnation temperatures and pressures are as follows respectively:

Scramjet Cruise:

$$T_{0,2,s} = 620.94 \text{ K}, P_{0,2,s} = 303234.69 \text{ Pa}$$

Ramjet Cruise:

$$T_{0,2,r} = 505.78 \text{ K}, P_{0,2,s} = 461187.02 \text{ Pa}$$

Since no stagnation losses were assumed in the diffuser design, these values were further taken as the inlet stagnation conditions for each design. This is consistent with the ideal isentropic diffuser assumption specified for the ramjet case and used here as a first order model for both operating points.

3.2.3. Static Flow Properties

Using the prescribed inlet and outlet Mach numbers, the corresponding static properties were calculated for both modes.

In both operating modes, the diffuser increases static pressure and temperature while reducing velocity, which is consistent with a compression and deceleration process, the ramjet case shows a strong reduction in velocity from 310.58 m/s to 110.2 m/s, while the scramjet mode reduces velocity from 618.27 m/s to 513.65 m/s without entering subsonic flow.

3.2.4. Diffuser Area Ratio

The required geometric change was obtained using the quasi-one-dimensional area-Mach correlation:

$$\frac{A}{A^*} = \left(\frac{1}{M}\right) * \left[\left(\frac{2}{\gamma+1}\right) * \left(1 + \left(\frac{\gamma-1}{2}\right) * M^2\right)\right]^{\frac{\gamma+1}{2(\gamma-1)}} \quad (26)$$

For the ramjet mode, the calculated values were $A_1/A^* = 1.06$ and $A_2/A^* = 2.34$, giving:

$$\frac{A_2}{A_1} = \frac{2.34}{1.06} = 2.21$$

For the scramjet mode, the calculated values were $A_1/A^* = 1.2$ and $A_2/A^* = 1.03$, giving:

$$\frac{A_2}{A_1} = \frac{1.03}{1.2} = 0.86$$

These results show that the ramjet diffuser must be diverging, while the scramjet diffuser is converging. This satisfies the quasi-one-dimensional isentropic flow theory, where subsonic deceleration requires an increase in area and supersonic deceleration requires a reduction in area.

3.2.5. Continuity Verification

The calculated area ratios were verified using the continuity equation:

$$\rho_1 V_1 A_1 = \rho_2 V_2 A_2 \text{ (6) and rearranged: } \frac{A_2}{A_1} = \frac{(rV)_1}{(rV)_2}$$

For Ramjet Conditions:

$$\frac{A_2}{A_1} = \frac{(rV)_1}{(rV)_2} = \frac{2.5 \times 320.58}{3.19 \times 110.2} = 2.21$$

For Scramjet conditions:

$$\frac{A_2}{A_1} = \frac{(rV)_1}{(rV)_2} = \frac{0.66 \times 618.27}{0.92 \times 513.65} = 0.86$$

The continuity results match the area-Mach solutions, confirming the diffuser sizing is internally consistent with the prerequisite assumptions.

3.2.6. Discussion

In Ramjet mode, the diffuser receives subsonic flow at $M_1 = 0.76$ and decelerates to $M_2 = 0.26$. the required ratio of $A_2/A_1 = 2.21$ shows that a diverging duct is required. This produced a static pressure rise from 321740.52 Pa to 441458.9 Pa together with an increase in static temperature from 465.45 K to 500.7 K. This is expected as, in subsonic isentropic flow, a reduction in velocity provides an increase in pressure.

In Scramjet mode, the diffuser receives a supersonic flow at $M_1 = 1.52$ and decelerates it to $M_2 = 1.2$ while remaining supersonic throughout. The area ratio required is $A_2/A_1 = 0.857$ – a converging duct requirement, with static pressure rising from 83521.98 Pa to 129937.17 Pa and static temperature increasing from 461.13 K to 510.64 K. This is again, consistent with the isentropic compressible-flow theory, where deceleration on the supersonic branch requires a reduction in cross sectional area.

A key design issue is that the two operating modes required an opposite area change: ramjet required a diverging diffuser whereas, scramjet required a converging diffuser. A fixed quasi-one-dimensional diffuser geometry cannot satisfy both requirements. Where in practical applications, this conflict in requirements is fixed using variable geometry, where the duct area is adjusted mechanically between operating modes. During ramjet operation the diffuser will expand to decelerate the subsonic flow and in scramjet operations the geometry contracts to decelerate the supersonic flow limiting to dropping into subsonic flow. This allows the same flow path to function across both operating modes without a fixed geometric compromise.

3.2.7. Limitations

This diffuser model is idealised as it assumes one-dimensional, adiabatic and isentropic flow with no total-pressure loss. Fluid behaviour such as viscosity effects and boundary-layer growth would reduce stagnation pressure and lower the pressure recovery at combustor entry.

These effects are especially important for high-speed internal compression systems, where the diffuser performance is sensitive to wall angle and surface losses. This result should therefore be interpreted as first order design values, with refinement including loss coefficients, and a more detailed geometric analysis.

3.3 Combustion Chamber

Assumptions

- The ratio of specific heats (γ) of the mix is constant through the chamber.
- The heat flux across the chamber is uniform.
- The inlet and diffuser are adiabatic. This means a constant stagnation temperature is maintained through the design up to the combustion chamber inlet.

As the flow across the engine inlet and the diffuser are adiabatic, the stagnation temperature at the combustion chamber inlet is equal to the freestream flow stagnation temperature ($T_{0, \text{freestream}}$). This is calculated by approximating the temperature of air at the appropriate altitudes. These temperatures ($T_{\text{Freestream}}$) were found by C. J. Brasefield [13] to be 213.15 K for 53,000 feet and 216.15 K at 65,500 feet. The value of 53,000 feet was used as an average as the aircraft increases in altitude from 40,500 to 65,500 feet over ramjet operation. Stagnation temperature was then calculated using Equation 27.

$$T_{0, \text{Freestream}} = T_{\text{Freestream}} * \left(1 + \left(\frac{\gamma - 1}{2} * M^2 \right) \right) \quad (27)$$

Therefore, by using the ratio of specific heats (γ) for air as 1.4 and the respective Mach numbers (M) for ramjet and scramjet air flow into the engine:

$$T_{0, \text{Ram freestream}} = 213.15 * \left(1 + \left(\frac{1.4 - 1}{2} * 2.62^2 \right) \right) \Rightarrow 505.779 \text{ K}$$

$$T_{0, \text{Scram freestream}} = 216.15 * \left(1 + \left(\frac{1.4 - 1}{2} * 3.02^2 \right) \right) \Rightarrow 620.938 \text{ K}$$

By using ram and scram Mach numbers into the chamber, a Rayleigh flow table was calculated with a ratio of specific heats value of 1.3 for the mix.

$$\frac{T_0}{T_0^*} = \frac{2Ma^2(\gamma + 1)}{(1 + \gamma Ma^2)^2} \left(1 + \frac{\gamma - 1}{2} Ma^2 \right) \quad (28)$$

$$\frac{T}{T^*} = \frac{(1 + \gamma)^2 Ma^2}{(1 + \gamma Ma^2)^2} \quad (29)$$

$$\frac{P}{P^*} = \frac{1 + \gamma}{1 + \gamma Ma^2} \quad (30)$$

$$\frac{P_0}{P_0^*} = \frac{(\gamma + 1)}{1 + \gamma Ma^2} \left(\frac{2 + (\gamma - 1)Ma^2}{\gamma + 1} \right)^{\gamma/\gamma-1} \quad (31)$$

$$\frac{\rho}{\rho^*} = \frac{u^*}{u} = \frac{1 + \gamma Ma^2}{(\gamma + 1)Ma^2} \quad (32)$$

Therefore, for chamber inlet:

Table 3: Mach Ratios for Chamber Inlet

Ma	T_0/T_0^*	T/T^*	P/P^*	P_0/P_0^*	$\rho/\rho^*=u^*/u$
0.26	0.265	0.302	2.114	1.205	6.997
1.2	0.977	0.924	0.801	1.020	0.867

As $T_{0, \text{Freestream}} = T_{01}$ in the combustion chamber as previously stated, the maximum stagnation temperature attainable (T_0^*) can be calculated.

$$T_{0, \text{Ram}}^* = 505.779 / \left(\frac{2 * 0.26^2 (1.3 + 1)}{(1 + 1.3 * 0.26^2)^2} \left(1 + \frac{1.3 - 1}{2} * 0.26^2 \right) \right) \Rightarrow 1905.623 \text{ K}$$

$$T_{0, \text{Scram}}^* = 620.938 / \left(\frac{2 * 1.2^2 (1.3 + 1)}{(1 + 1.3 * 1.2^2)^2} \left(1 + \frac{1.3 - 1}{2} * 1.2^2 \right) \right) \Rightarrow 635.863 \text{ K}$$

To then calculate the outlet Mach number, it was identified that T_{02}/T_{02}^* is constrained due to Rayleigh choking. As heat is added, the Mach number moves to 1. Therefore, as the inlet Mach for ram and scram are 0.26 Mach and 1.2 Mach respectively, the range that the outlet Mach number is limited to is between 0.26 to 1 for ram and 1.2 to 1 for scram. This then limits the outlet stagnation temperature, T_{02} , to between the inlet stagnation temperature, T_{01} , for no increase in Mach number and T_0^* , the maximum stagnation temperature possible, indicating maximum possible heat addition. Therefore, the ranges were between 505.779 and 1905.623 K for ram and 620.938 and 635.863 K for scram. These ranges were used in the calculations for the CD nozzle

to optimize the design conditions and ensure that Rayleigh choking would not occur. This small range for the scramjet condition means that the allowable heat addition is small also when compared to the heat addition required for thrust requirements to be met.

The outlet stagnation temperatures which were identified to be used were 1700 K for ram and 628.380 K for scram. As these are within the specified ranges, the Mach number does not reach 1 and therefore Rayleigh choking does not occur. The Mach number at the outlet was then calculated by the table created previously.

$$\frac{T_{02}}{T_0^*} = \frac{1700}{1905.623} = 0.892$$

The Mach number corresponding to this value is 0.694. This is therefore outlet Mach number of the chamber for ramjet configuration.

$$\frac{T_{02}}{T_0^*} = \frac{628.380}{635.863} = 0.988$$

The Mach number corresponding to this value is 1.135. This is therefore outlet Mach number of the chamber for scramjet configuration.

To calculate the specific heat capacity of the mixture (c_p), the ratio of heat capacities (γ) and specific gas constant (R) of $276 \text{ J kg}^{-1} \text{ K}^{-1}$ was used.

$$c_p = \frac{\gamma R}{\gamma - 1} \quad (33)$$

$$c_p = \frac{1.3 * 276}{1.3 - 1} \Rightarrow 1196 \text{ J kg}^{-1} \text{ K}^{-1}$$

To find the mass flow rate of the fuel ($\dot{m}_{\text{fuel}}$) into the chamber, energy balance was used. The total mass flowrate of the air and fuel mix ($\dot{m}_{\text{total}}$) required by the CD nozzle of 42.1 kg s^{-1} for ram and 57.77 kg s^{-1} was used.

$$\dot{m}_{\text{total}} c_p T_{02} = \dot{m}_{\text{air}} c_p T_{01} + \dot{m}_{\text{fuel}} q_{\text{fuel}}$$

$$\dot{m}_{\text{total}} c_p T_{02} = (\dot{m}_{\text{total}} - \dot{m}_{\text{fuel}}) c_p T_{01} + \dot{m}_{\text{fuel}} q_{\text{fuel}}$$

Expanding and simplifying this in terms of $\dot{m}_{\text{fuel}}$;

$$\dot{m}_{\text{fuel}} = \frac{\dot{m}_{\text{total}} c_p (T_{02} - T_{01})}{q_{\text{fuel}} - c_p T_{01}} \quad (34)$$

$$\dot{m}_{fuel,ram} = \frac{42.1 * 1196 * (1700 - 505.779)}{(43.5 * 10^6) - (1196 * 505.779)} \Rightarrow 1.385 \text{ kgs}^{-1}$$

$$\dot{m}_{fuel,scram} = \frac{57.77 * 1196 * (628.380 - 620.938)}{(43.5 * 10^6) - (1196 * 620.938)} \Rightarrow 0.0120 \text{ kgs}^{-1}$$

The total heat transferred into the system (Q) can be calculated.

$$Q = \dot{m}_{total} c_p (T_{02} - T_{01}) \quad (35)$$

$$Q_{ram} = 42.1 * 1196 * (1700 - 505.779) \Rightarrow 60.131 \text{ MJ}$$

$$Q_{scram} = 57.77 * 1196 * (628.380 - 620.938) \Rightarrow 514.051 \text{ kJ}$$

Maximum allowable heat transfer (Q_{lc}) can be calculated as T_0^* is the maximum possible temperature across the chamber.

$$Q_{lc} = c_p (T_0^* - T_{0,1}) \quad (36)$$

$$Q_{lc,Ram} = 42.1 * 1196 * (1905.623 - 505.779) \Rightarrow 70.484 \text{ MJ}$$

$$Q_{lc,Scram} = 57.77 * 1196 * (635.863 - 620.938) \Rightarrow 1.031 \text{ MJ}$$

As $Q < Q_{lc}$, this heat transfer is possible without causing Rayleigh choking.

The heat added per unit length (Q/L) was calculated. This was done by assuming uniform heat flux. M^{-1}

$$\frac{Q_{ram}}{L} = \frac{60.131 * 10^6}{0.51} = 117.904 \text{ MJm}^{-1}$$

$$\frac{Q_{scram}}{L} = \frac{514.051 * 10^3}{0.51} = 1.008 \text{ MJm}^{-1}$$

The static temperature of the flow at the inlet (T_1) and outlet (T_2) can be calculated.

$$T = \frac{T_0}{1 + (\frac{\gamma - 1}{2} M^2)} \quad (37)$$

Therefore,

$$\begin{aligned}
 T_{1,ram} &= \frac{T_{01,Ram}}{1 + \left(\frac{\gamma-1}{2} M_{1,Ram}^2\right)} \Rightarrow \frac{505.779}{1 + \left(\frac{1.3-1}{2} * 0.26^2\right)} \Rightarrow 500.702 \text{ K} \\
 T_{1,Scram} &= \frac{T_{01,Scram}}{1 + \left(\frac{\gamma-1}{2} M_{1,Scram}^2\right)} \Rightarrow \frac{620.938}{1 + \left(\frac{1.3-1}{2} * 1.2^2\right)} \Rightarrow 510.640 \text{ K} \\
 T_{2,Ram} &= \frac{T_{02,Ram}}{1 + \left(\frac{\gamma-1}{2} M_{2,Ram}^2\right)} \Rightarrow \frac{1700}{1 + \left(\frac{1.3-1}{2} * 0.69^2\right)} \Rightarrow 1586.687 \text{ K} \\
 T_{2,Scram} &= \frac{T_{02,Scram}}{1 + \left(\frac{\gamma-1}{2} M_{2,Scram}^2\right)} \Rightarrow \frac{628.380}{1 + \left(\frac{1.3-1}{2} * 1.135^2\right)} \Rightarrow 526.619 \text{ K}
 \end{aligned}$$

Due to the high outlet temperature in the ramjet configuration, the engine must use advanced materials or thermal barrier coatings to avoid deformation of the chamber. The scramjet condition does not experience this high temperature, however.

The stagnation pressure which the nozzle requires (P_{02}) is 400 kPa for ram and 300 kPa for scram. From this, the static pressure at the outlet (P_2) can be calculated.

$$P = P_0 \left(1 + \frac{\gamma-1}{2} M^2\right)^{-\frac{\gamma}{\gamma-1}} \quad (38)$$

Therefore,

$$\begin{aligned}
 P_{2,Ram} &= P_{02,Ram} \left(1 + \frac{\gamma-1}{2} M_{2,Ram}^2\right)^{-\frac{\gamma}{\gamma-1}} \Rightarrow 400 * 10^3 * \left(1 + \frac{1.3-1}{2} * 0.69^2\right)^{-\frac{1.3}{1.3-1}} \\
 &= 296.649 \text{ kPa} \\
 P_{2,Scram} &\Rightarrow P_{02,Scram} \left(1 + \frac{\gamma-1}{2} M_{2,Scram}^2\right)^{-\frac{\gamma}{\gamma-1}} = 300 * 10^3 * \left(1 + \frac{1.3-1}{2} * 1.135^2\right)^{-\frac{1.3}{1.3-1}} = \\
 &> 139.522 \text{ kPa}
 \end{aligned}$$

Then, the static pressure at the inlet of the chamber can be calculated.

$$P_1 = P_2 * \frac{1 + \gamma M_2^2}{1 + \gamma M_1^2} \quad (39)$$

Therefore,

$$P_{1,Ram} = P_{2,Ram} * \frac{1 + \gamma M_{2,Ram}^2}{1 + \gamma M_{1,Ram}^2} \Rightarrow 296.649 * 10^3 * \frac{1 + (1.3 * 0.69^2)}{1 + (1.3 * 0.26^2)} \Rightarrow 441.459 \text{ kPa}$$

$$P_{1,Scram} = P_{2,Scram} * \frac{1 + \gamma M_{2,Scram}^2}{1 + \gamma M_{1,Scram}^2} \Rightarrow 139.522 * 10^3 * \frac{1 + (1.3 * 1.135^2)}{1 + (1.3 * 1.2^2)} \Rightarrow 129.937 \text{ kPa}$$

The stagnation pressure at the inlet (P_{01}) can then be calculated.

$$P_0 = P \left(1 + \frac{\gamma - 1}{2} M^2 \right)^{\frac{\gamma}{\gamma - 1}} \quad (40)$$

Therefore,

$$P_{01,Ram} = P_{1,Ram} \left(1 + \frac{\gamma - 1}{2} M_{1,Ram}^2 \right)^{\frac{\gamma}{\gamma - 1}} \Rightarrow 441.459 * 10^3 * \left(1 + \frac{1.3 - 1}{2} * 0.26^2 \right)^{\frac{1.3}{1.3 - 1}} \\ = 461.187 \text{ kPa}$$

$$P_{01,Scram} = P_{1,Scram} \left(1 + \frac{\gamma - 1}{2} M_{1,Scram}^2 \right)^{\frac{\gamma}{\gamma - 1}} \Rightarrow 129.937 * 10^3 * \left(1 + \frac{1.3 - 1}{2} * 1.2^2 \right)^{\frac{1.3}{1.3 - 1}} \\ = 303.234 \text{ kPa}$$

The density (ρ) and velocity (u) of the flow at the inlet can be used to calculate the cross-sectional area (A) of the chamber.

$$\rho = \frac{P_1}{RT_1} \quad (41)$$

$$\rho_{1,ram} = \frac{441459}{276 * 500.702} \Rightarrow 3.194$$

$$\rho_{1,scram} = \frac{129937}{276 * 510.640} \Rightarrow 0.922$$

$$u = M_1 \sqrt{\gamma R T_1} \quad (42)$$

$$u_{1,ram} = 0.26 * \sqrt{1.3 * 276 * 500.702} \Rightarrow 110.202$$

$$u_{1,scram} = 1.2 * \sqrt{1.3 * 276 * 510.640} \Rightarrow 513.647$$

The cross-sectional area can then be calculated.

$$A = \frac{\dot{m}_{air}}{\rho u} \quad (43)$$

$$A_{ram} = \frac{42.1 - 1.385}{3.194 * 110.202} \Rightarrow 0.116m^2$$

$$A_{scram} = \frac{57.77 - 0.012}{0.922 * 513.647} \Rightarrow 0.122m^2$$

As the geometry of the chamber should be equal in both flow conditions, the discrepancy of ~5.17% likely stems from the fact that the mass flow rates and outlet pressures have been set by the nozzle rather than being optimized for the single duct area. In a real system, the area would therefore likely need to be designed to the average area of 0.119m².

The outlet Mach number and mass flowrate of fuel required can be graphed to determine the maximum mass flowrate of fuel before choking occurs.

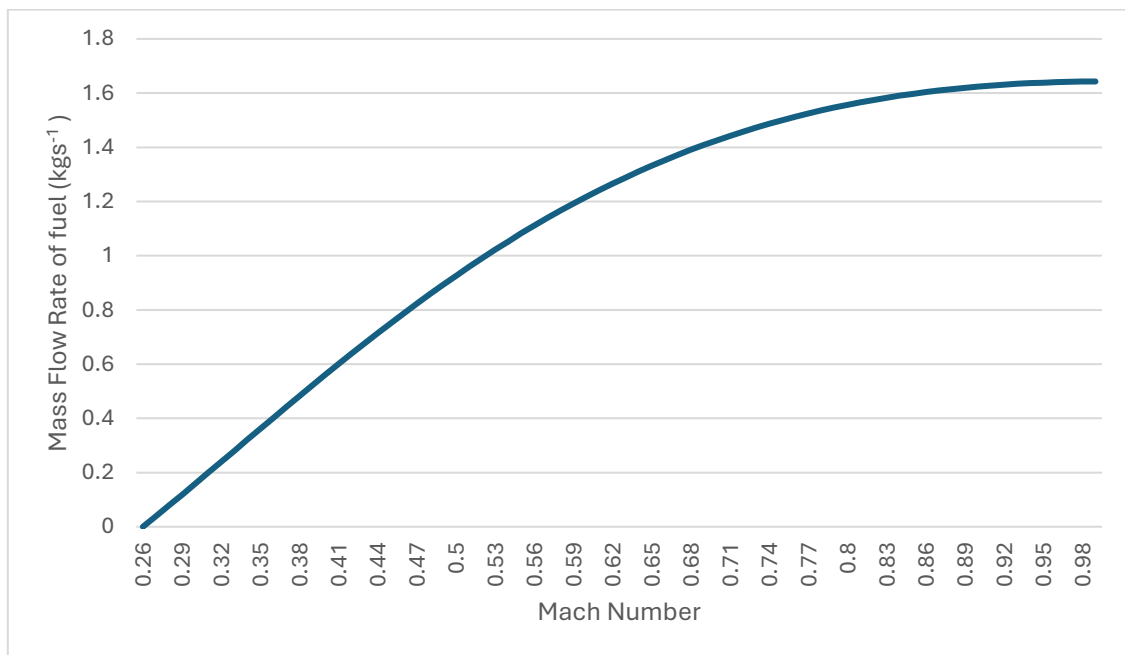


Figure 15: RAM Mass Flow Rate Across Outlet Mach Range

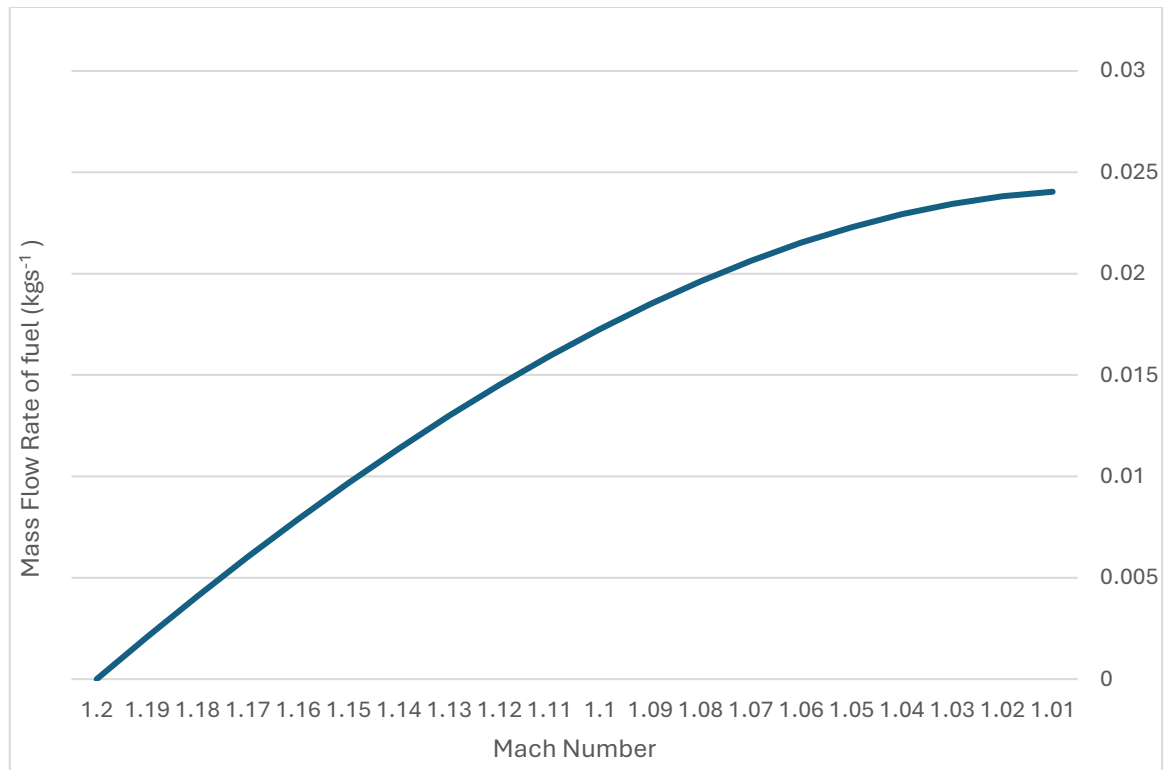


Figure 16: SCRAM Mass Flow Rate Across Outlet Mach Range

The graphed values follow the expected relationship. The mass flow rate starts at 0 as to maintain the Mach number at the inlet, no heat addition is required. The relationship then increases across the Mach sweep and levels once a value of 1 is reached, indicating choked flow.

The final calculated values can be seen below.

Table 4: Combustion Chamber Final Parameters

	Parameter	Symbol	Ram	Scram	Units
Inlet Conditions					
	Mach	M1	0.26	1.2	
	Stagnation Temperature	T01	505.779	620.938	K
	Static Temperature	T1	500.70188	510.639803	K
	Stagnation Pressure	P01	461187.02	303234.688	Pa
	Static Pressure	P1	441458.9	129937.165	Pa
	Density	p1	3.194	0.922	kgm ⁻³
	Velocity	u1	110.202	513.647	ms ⁻¹
	Cross-sectional Area	A	0.116	0.122	m ²
Outlet Conditions					
	Mach	M2	0.69	1.135	
	Stagnation Temperature	T02	1700	628.38	K
	Static Temperature	T2	1.59E+03	5.27E+02	K
	Stagnation Pressure	P02	400000	300000	Pa
	Static Pressure	P2	2.97E+05	1.40E+05	Pa
	Density	p2	6.77E-01	9.60E-01	kgm ⁻³

	Parameter	Symbol	Ram	Scram	Units
	Velocity	u_2	5.21E+02	4.93E+02	ms^{-1}
	Cross-sectional Area	A	1.15E-01	1.22E-01	m^2
Mass Flow					
	Mass flow rate of fuel	$\dot{m}_{\text{fuel}}$	1.385	0.012	kgs^{-1}
	Mass flow rate of mix	$\dot{m}_{\text{mix}}$	42.1	57.77	kgs^{-1}
Mix Properties					
	Heat capacity ratio	γ	1.3	1.3	
	Specific heat capacity	cp	1196	1196	$\text{Jkg}^{-1}\text{K}^{-1}$
	Gas constant	r	276	276	$\text{Jkg}^{-1}\text{K}^{-1}$
Heat Transferred					
	Heat transferred	Q	60.131	0.514	MJ
	Maximum heat transfer	Q_{lc}	70.484	1.031	MJ

3.3. Converging-Diverging Nozzle

The converging–diverging nozzle converts combustor stagnation pressure and temperature into a high-momentum exhaust jet, setting the engine exit velocity, discharge pressure and net thrust. For this DMRJ design, the nozzle must provide enough momentum to accelerate the 64,500 kg aircraft from Mach 2.62 to Mach 3.02 in 30 minutes, while also achieving 30.3 kN thrust at scramjet cruise. The analysis therefore considers choking, expansion state, off-design shock behaviour, mass flow rate, force and minimum operating altitude. A quasi-one-dimensional isentropic model was adopted because the brief permits this approach for CD nozzle design. Sample calculations may be found in the appendix.

Table 5: Design Assumptions.

Category	Parameter	Value	Unit
Design assumption	Required thrust per engine	30.3	kN
Preliminary cruise nozzle-inlet state	$(P_{0,5})$	300	kPa
Preliminary cruise nozzle-inlet state	$(T_{0,5})$	1800	K
Off-design ramjet check	$(P_{0,5,r})$	400	kPa
Off-design ramjet check	$(T_{0,5,r})$	1700	K

3.3.1. Freestream conditions

$$T_{\infty} = 216.65K$$

Atmospheric speed of sound may be found by:

$$a_{\infty} = \sqrt{\gamma_a R_a T_{\infty}} \quad (44)$$

With

$$\gamma_a = 1.4, R_a = 287 Jkg^{-1}K^{-1}$$

Therefore, the freestream velocities can be found by:

$$V_{\infty} = M_{\infty,s} a_{\infty} \quad (45)$$

Using ISA, the ambient pressures are approximately,

$$P_{\infty,s} = 5.51 kPa$$

$$P_{\infty,r} = 18.31 kPa$$

3.3.2. Minimum Acceleration Force

The 64,500kg aircraft must accelerate from Mach 2.62 to 3.02 in 30 minutes.

Therefore, the minimum horizontal force is:

$$F_{acc} = m \frac{\Delta V}{\Delta t} \quad (46)$$

with

$$\Delta V = V_{\infty,s} - V_{\infty,r} \quad (47)$$

Since the aircraft also requires a cruise force of 30.3kN per engine, this is the governing design force.

3.3.3. Design Exit Mach Number at Cruise

For a first-pass design, the nozzle was assumed to be ideally expanded at cruise.

Therefore,

$$P_e = P_{\infty,s} = 5.51 \text{ kPa} \quad (48)$$

The isentropic pressure relation for the combusting mixture is

$$\frac{P_0}{P} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{\frac{\gamma - 1}{\gamma}} \quad (49)$$

Rearranging for the exit Mach number,

$$M_e = \sqrt{\frac{2}{\gamma - 1} \left[\left(\frac{P_0}{P_e}\right)^{\frac{\gamma - 1}{\gamma}} - 1 \right]} \quad (50)$$

3.3.4. Nozzle Area Ratio

The area-Mach relationship is:

$$\frac{A_e}{A^*} = \frac{1}{M_e} \left[\frac{2}{\gamma + 1} \left(1 + \frac{\gamma - 1}{2} M_e^2\right) \right]^{\frac{\gamma + 1}{2(\gamma - 1)}} \quad (51)$$

3.3.5. Exit Temperature

The isentropic relationship is:

$$\frac{T_0}{T_e} = 1 + \frac{\gamma - 1}{2} M_e^2 \quad (52)$$

Hence,

$$T_e = \frac{T_0}{1 + \frac{\gamma - 1}{2} M_e^2} \quad (53)$$

3.3.6. Exit Velocity

The speed of sound is calculated by:

$$a_e = \sqrt{\gamma R T_e} \quad (54)$$

Therefore, the exhaust velocity is as follows.

$$V_e = M_e a_e \quad (55)$$

3.3.7. Required Mass Flow Rate

Nozzle thrust is determined using the following.

$$F = \dot{m}(V_e - V_\infty) + (P_e - P_\infty)A_e \quad (56)$$

At the cruise design point, the ideal expansion means

$$P_e = P_\infty \quad (57)$$

So, the pressure-thrust term is zero and

$$F = \dot{m}(V_e - V_\infty) \quad (58)$$

Therefore,

$$\dot{m}_{scramjet} = \frac{F}{V_e - V_\infty} \quad (59)$$

Since both P_0 and T_0 are preliminary combustor estimates, a parametric sensitivity study was carried out to quantify the dependence of $\dot{m}$ on these inputs. With the throat area fixed by geometry, the choked-flow equation gives $\dot{m} \propto P_0 / \sqrt{T_0}$, so mass flow rate scales linearly with inlet stagnation pressure and only weakly with inlet stagnation temperature. Tables 1 and 2 present the computed mass flow rates over a $\pm 20\%$ range of P_0 and a ± 200 K range of T_0 for the scramjet and ramjet modes respectively, with the design points highlighted.

Table 6, Mixture mass flow rate(kg/s)- Scramjet mode. Design point highlighted in amber.

T_0 (K) \ P_0 (kPa)	240	260	280	300	320	340	360
1600	35.73	38.71	41.69	44.66	47.64	50.62	53.60
1700	34.66	37.55	40.44	43.33	46.22	49.11	52.00
1800	33.69	36.50	39.30	42.11	44.92	47.72	50.53
1900	32.79	35.52	38.25	40.99	43.72	46.45	49.18
2000	31.96	34.62	37.29	39.95	42.61	45.28	47.94

Table 7, Mixture mass flow rate(kg/s)- Ramjet mode. Design point highlighted in amber.

T_o (K) \ P_o (kPa)	320	350	380	400	420	450	480
1500	49.20	53.82	58.43	61.51	64.58	69.19	73.81
1600	47.64	52.11	56.58	59.55	62.53	67.00	71.46
1700	46.22	50.55	54.89	57.77	60.66	65.00	69.33
1800	44.92	49.13	53.34	56.15	58.95	63.17	67.38
1900	43.72	47.82	51.92	54.65	57.38	61.48	65.58

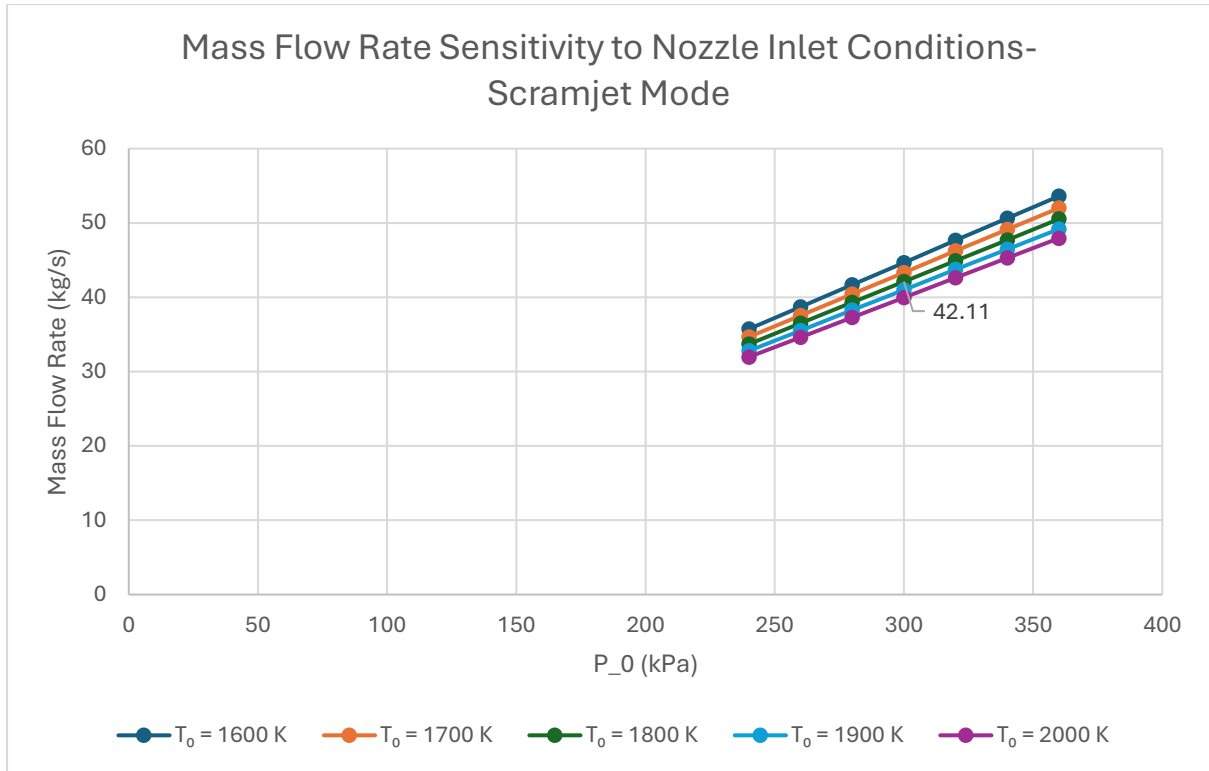


Figure 17: Mass Flow Rate Sensitivity to Nozzle Inlet Conditions- Scramjet Mode.

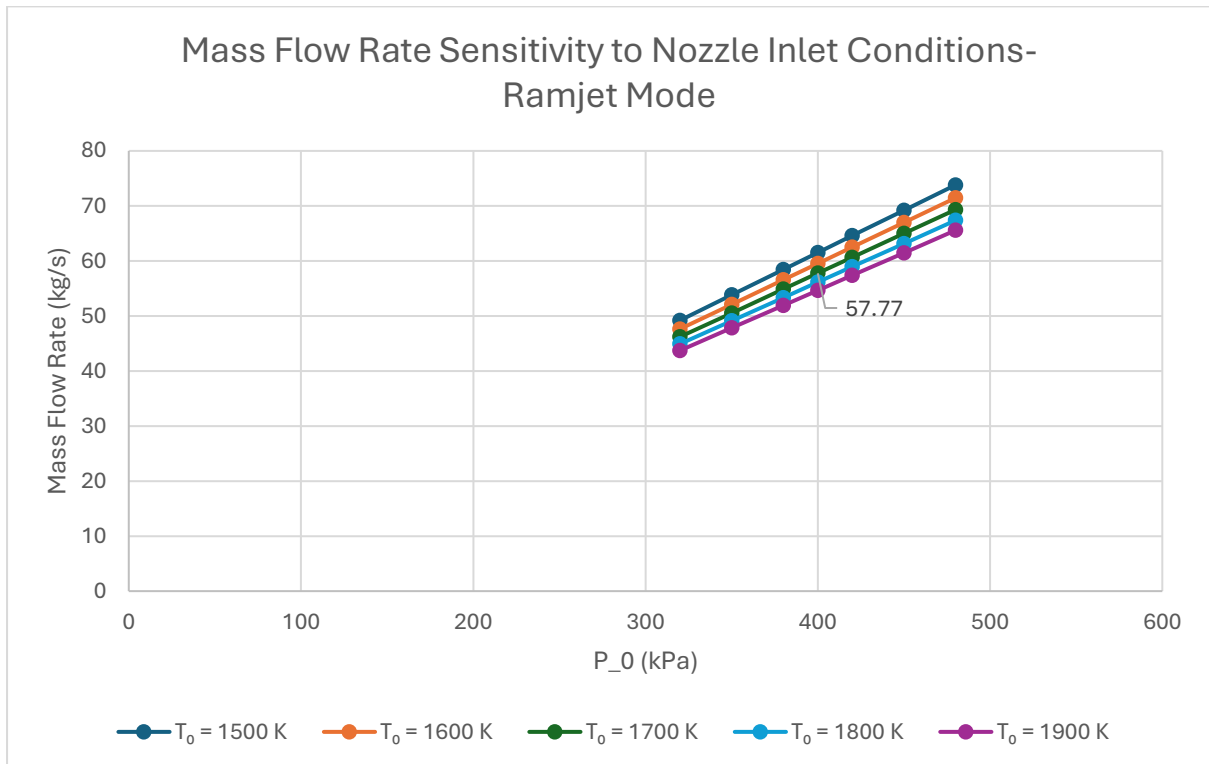


Figure 18: Mass Flow Rate Sensitivity to Nozzle Inlet Conditions- Ramjet Mode.

The sensitivity study shows that mass flow rate is far more affected by stagnation pressure than stagnation temperature. In scramjet mode, reducing P_0 by 20% from 300 kPa to 240 kPa lowers $\dot{m}$ by about 8.4 kg s^{-1} , also 20%, while increasing T_0 by 200 K reduces $\dot{m}$ by only 1.5 kg s^{-1} , or 3.6%.

This agrees with the relationship $\dot{m} \propto P_0/\sqrt{T_0}$. Since thrust scales with mass flow rate, accurate combustor outlet pressure is critical for reliable thrust prediction.

3.3.8. Throat Area from Choking

The choked mass-flow equation is:

$$\dot{m} = A^* \frac{P_0}{\sqrt{T_0}} \sqrt{\frac{\gamma}{R}} \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma+1}{2(\gamma-1)}} \quad (60)$$

This is rearranged for throat area.

Then

$$A_e = \left(\frac{A_e}{A^*} \right) A^* \quad (61)$$

3.3.9. Circular Diameters

Assuming a circular throat and exit:

$$d_t = \sqrt{\frac{4A^*}{\pi}} \quad (62)$$

$$d_e = \sqrt{\frac{4A_e}{\pi}} \quad (63)$$

3.3.10. Validation

The validation equations for choking, off-design ramjet and normal-shock check are given below.

3.3.1.1 Choking Check

The critical pressure ratio for choking is:

$$\frac{P^*}{P_0} = \left(\frac{2}{\gamma + 1} \right)^{\frac{1.3}{0.3}} \quad (64)$$

$$\therefore \frac{P^*}{P_0} = 0.5457$$

The nozzle is choked if,

$$\frac{P_b}{P_0} \leq 0.5457$$

At cruise,

$$\frac{P_{\infty,s}}{P_{0,5}} = \frac{5.51}{300} = 0.0184$$

$$0.0184 < 0.5457$$

$\therefore$ the nozzle is choked at cruise.

At the ramjet point, using the assumed off-design total pressure, 400kPa:

$$\frac{P_{\infty,r}}{P_{0,5,r}} = \frac{18.31}{400} = 0.0458$$

$$0.0485 < 0.5457$$

$\therefore$ the nozzle is also choked at the ramjet point.

3.3.1.2 Off-design Ramjet Check

The area ratio remains the same at 6.32 for the supersonic branch, meaning the exit Mach number remains approximately 3.18.

Using the assumed ramjet nozzle-inlet state, the exit temperature is:

$$\begin{aligned} T_{e,r} &= \frac{1700}{1 + 0.15(3.178)^2} \\ &= 675.83K \end{aligned}$$

The exit pressure is:

$$\begin{aligned} P_{e,r} &= \frac{400000}{(1 + 0.15(3.178)^2)^{\frac{1.3}{0.3}}} \\ &= 7.35 \text{ kPa} \end{aligned}$$

The exit velocity is:

$$\begin{aligned} V_{e,r} &= M_{e,r} \sqrt{\gamma R T_{e,r}} \\ &= 3.178 \sqrt{1.3 \times 276 \times 675.83} \end{aligned} \tag{65}$$

$$\therefore V_{e,r} = 1565.18 \text{ ms}^{-1}$$

The choked mass flow rate at the ramjet point becomes:

$$\dot{m}_r = A^* \frac{P_{0,5,r}}{\sqrt{T_{0,5,r}}} \sqrt{\frac{\gamma}{R} \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma+1}{2(\gamma-1)}}} \tag{66}$$

$$\therefore \dot{m}_r = 57.78 \text{ kg s}^{-1}$$

The thrust is then,

$$\begin{aligned} F_r &= \dot{m}_r (V_{e,r} - V_{\infty,r}) + ((P_{e,r} - P_{\infty,r}) A_e) \\ F_r &= 57.78(1565.18 - 773.01) + (7.35 - 18.31) \times 10^3(0.9372) \\ &= 35.49 \text{ kN} \end{aligned}$$

Since

$$P_{e,r} = 7.35 \text{ kPa} < P_{\infty,r} = 18.31 \text{ kPa}$$

The nozzle is over-expanded in ramjet mode.

3.3.1.3 Normal-shock Check

For the supersonic exit Mach number $M_1 = 3.178$, the normal-shock static pressure rise is:

$$\begin{aligned} \frac{P_2}{P_1} &= 1 + \frac{2\gamma}{\gamma + 1} (M_1^2 - 1) \\ &= 1 + \frac{2(1.3)}{2.3} (3.178^2 - 1) \end{aligned} \quad (67)$$

$$\therefore \frac{P_2}{P_1} = 11.29$$

Before the shock, the exit static-to-stagnation pressure ratio is:

$$\begin{aligned} \frac{P_1}{P_0} &= \frac{1}{\left(1 + \frac{\gamma - 1}{2} M_1^2\right)^{\frac{\gamma}{\gamma - 1}}} \\ &= \frac{1}{54.446} \end{aligned} \quad (68)$$

$$\therefore \frac{P_1}{P_0} = 0.01837$$

Therefore, the back-pressure ratio for a shock at the nozzle exit is:

$$\frac{P_b}{P_0} = \frac{P_2}{P_1} \frac{P_1}{P_0} \quad (69)$$

$$= 11.29 \times 0.01837$$

$$\therefore \frac{P_b}{P_0} = 0.207$$

So, a normal shock stays outside the nozzle.

$$\frac{P_{\infty}}{P_0} = < 0.207$$

At cruise:

$$\frac{5.51}{300} = 0.0184 < 0.207$$

At the ramjet point:

$$\frac{18.31}{400} = 0.0458 < 0.207$$

$$\rho_{in,s} = \frac{268602.3}{276 * 1455.21} = 0.669 \text{ kg/m}^3$$

$$V_{2,rs} = 0.26 * 408.29 = 106.16 \text{ m/s}$$

3.3.11. Engine Flow Property Profile and Expansion Status

To check that the converging-diverging nozzle calculation behaves consistently across both operating modes, the calculated stagnation and static temperatures were plotted station-by-station. The profile was generated from the preliminary nozzle-inlet states, the isentropic expansion relations used above, and the fixed nozzle geometry. This provides a visual check that the nozzle is converting thermal energy into exhaust kinetic energy, rather than relying only on the final numerical thrust value.

Table 8: Engine flow property profile used for station-by-station comparison.

Station	Physical location	T0 Scramjet (K)	T Scramjet (K)	T0 Ramjet (K)	T Ramjet (K)
0	Ambient / freestream	216.65	216.65	216.65	216.65
1	Nozzle inlet	1800	1800	1700	1700
2	Throat	1800	1565.22	1700	1478.26
3	Nozzle exit	1800	775.59	1700	675.83

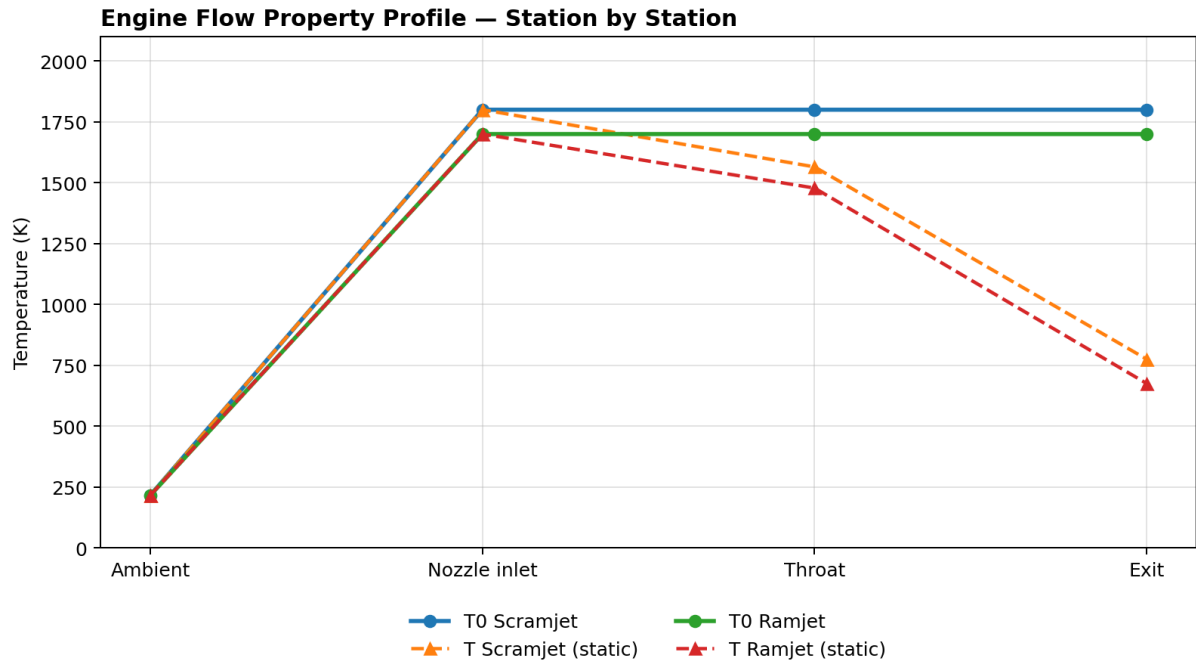


Figure 19: Engine flow property profile - station by station.

Figure 3 shows temperature rising from ambient to the nozzle inlet due to upstream heat addition. Stagnation temperature remains nearly constant through the nozzle, while static temperature falls sharply from throat to exit, confirming conversion of static enthalpy into kinetic energy. The scramjet curve is slightly higher due to its hotter nozzle-inlet condition.

Table 9: Exit velocity and expansion-status comparison.

Operating mode	Flight Mach number	Exit velocity (m/s)	Freestream velocity (m/s)	Expansion interpretation
Scramjet cruise	3.02	1610.56	891.03	Ideally expanded at design cruise
Ramjet acceleration	2.62	1565.18	773.01	Over-expanded off-design condition

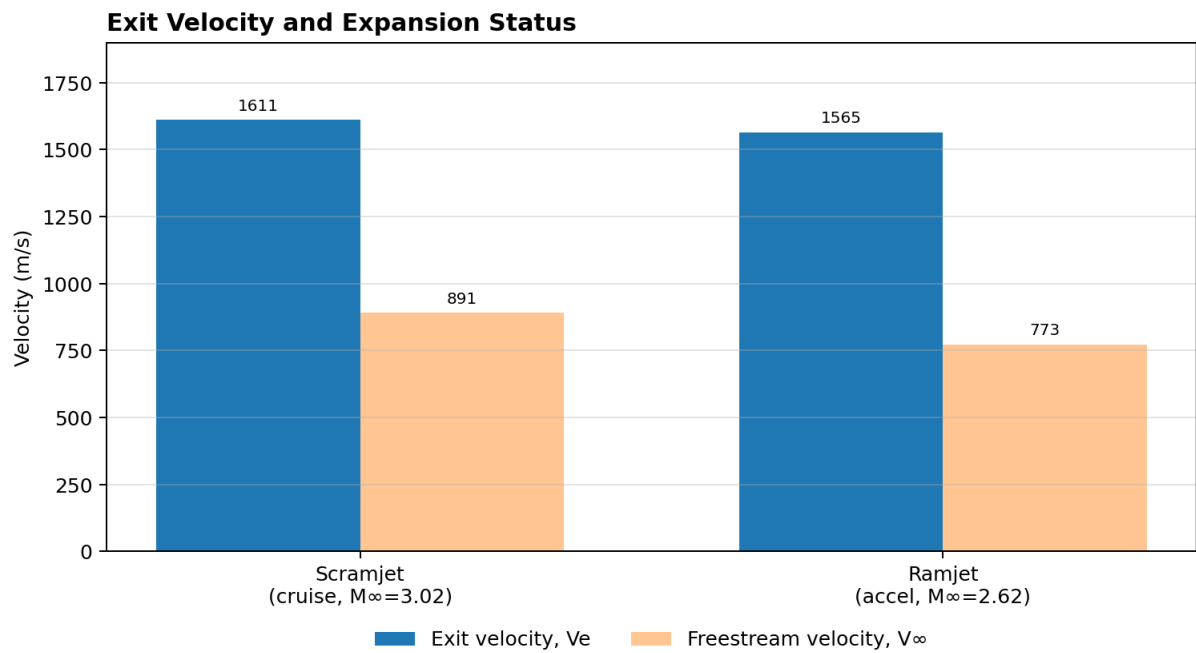


Figure 20: Exit velocity and expansion status comparison.

Figure 4 shows that both modes produce exhaust velocities above flight velocity, giving positive momentum thrust. The scramjet case has the higher exit velocity due to its higher Mach number and inlet energy. However, the ramjet case is over-expanded, so the fixed nozzle is best matched to scramjet cruise.

3.3.12. Minimum Altitude to Avoid Internal Discontinuities

$$P_{\infty, max} = 0.207 P_0 \quad (70)$$

For the cruise,

$$P_{\infty, max} = 0.207(300) = 62.1 \text{ kPa}$$

Which corresponds to an altitude of roughly

$$h_{min} \approx 3.9 \text{ km}$$

For the ramjet,

$$P_{\infty, max} = 0.207(400) = 82.8 \text{ kPa}$$

Which corresponds to an altitude of roughly

$$h_{min} \approx 1.7 \text{ km}$$

Both design operating points are comfortably above the shock-free minimum altitude.

3.3.13. Results and Discussion

Table 10: Results Summary.

Category	Parameter	Symbol / Unit	Scramjet Mode (Cruise)	Ramjet Mode (Acceleration)
Flight Conditions	Flight Mach Number	M_∞ (-)	3.02	2.62
	Altitude	h (ft)	65,500	40,500
	Ambient Static Pressure	P_∞ (Pa)	5,504	18,308
	Ambient Static Temperature	T_∞ (K)	216.65	216.65
	Flight Velocity	V_∞ (m/s)	891.03	773.01
Design Target	Required Thrust	F_{required} (N)	30,300	30,300
	Aircraft Mass	m (kg)	64,500	64,500
	Acceleration-Only Force	F_{acc} (N)	4,229	4,229
Combustor Exit (Nozzle Inlet)	Stagnation Temperature	$T_{0,5}$ (K)	1,800	1,700
	Stagnation Pressure	$P_{0,5}$ (Pa)	300,000	400,000
	Mixture Specific Heat Ratio	γ (-)	1.3	1.3
	Mixture Gas Constant	R (J/kgK)	276	276
Nozzle Geometry (Fixed)	Throat Area	A^* (m ²)	0.1482	0.1482
	Exit Area	A_e (m ²)	0.9377	0.9377
	Area Expansion Ratio	A_e / A^* (-)	6.33	6.33
	Throat Diameter	d_t (m)	0.4345	0.4345
	Exit Diameter	d_e (m)	1.0927	1.0927
	Exit Mach Number	M_e (-)	3.179	3.179
Nozzle Exit Conditions	Exit Static Pressure	P_e (Pa)	5,504	7,339
	Expansion Status	-	Perfectly Expanded	Over-expanded
	Exit Static Temperature	T_e (K)	715.42	675.67
	Exit Velocity	V_e (m/s)	1610.69	1565.31
	Mass Flow Rate	$\dot{m}$ (kg/s)	42.10	57.77
	Net Thrust	F (N)	30,300	35,481
Shock / Off-Design Check	Back-pressure ratio for normal shock at nozzle exit	$(p_b/p_0)_{\text{exit shock}}$ (-)	0.2072	0.2072
	Minimum total pressure to avoid internal normal shock	$P_{0,\text{min}}$ (kPa)	26.6	88.3
	Minimum altitude to avoid internal normal shock	h_{min} (km)	3.93	1.66

The converging–diverging nozzle model produced a reasonable first-pass design for the scramjet cruise condition. As the nozzle was assumed to be ideally expanded at cruise, the exit pressure matched ambient pressure, and the pressure-thrust term was zero. Thrust was therefore generated mainly through momentum change. The 30.3 kN cruise thrust requirement governed the design, rather than the smaller acceleration force, indicating that the nozzle is sized primarily for sustained high-speed cruise.

The calculated exit Mach number of approximately 3.18 and the large nozzle area ratio show that strong supersonic expansion is required. This is physically sensible because the aircraft is already travelling at high freestream velocity, so the exhaust must be accelerated well above flight speed to generate useful net thrust. However, this makes the result highly dependent on the assumed combustor outlet stagnation conditions.

The sensitivity study showed that mass flow rate is much more sensitive to stagnation pressure than stagnation temperature. A 20% reduction in P_0 caused an approximately 20% reduction in

mass flow rate, while a 200 K change in T_0 had a much smaller effect. Since P_0 was estimated, any inlet, isolator or combustor pressure loss would directly reduce mass flow rate and thrust, so the predicted performance may be optimistic.

The off-design ramjet case remained choked but was no longer ideally expanded. Its exit pressure was below ambient pressure, indicating over-expansion and a negative pressure-thrust contribution. Although no internal normal shock was predicted, this shows that the cruise-optimised fixed geometry is not fully suitable for ramjet operation. A variable-geometry nozzle would likely improve dual-mode performance.

Overall, the model is useful for preliminary sizing and confirms expected energy conversion, but the results should not be treated as final design values. Further refinement should include pressure losses, variable gas properties and improved off-design modelling.

3.3.14. Model Limitations

Table 11: Model Limitations.

Limitation	Effect on nozzle prediction	Likely impact
One-dimensional, steady, isentropic flow assumed	Neglects wall friction, boundary-layer growth, turbulence, heat transfer and viscous losses	Exit velocity and thrust are likely overestimated
Combustor outlet (P_0) and (T_0) are estimated	Nozzle performance depends strongly on the assumed inlet stagnation state	Errors in (P_0) directly affect mass flow rate and thrust
Constant (γ) and (R) used	Real combustion products vary with temperature and composition	Exit Mach number, speed of sound, temperature and velocity may differ
Fixed nozzle geometry used for both modes	Geometry is optimised for scramjet cruise, not ramjet operation	Ramjet case becomes over-expanded, reducing performance
No detailed shock/boundary-layer interaction modelling	Off-design behaviour is assessed using simplified normal-shock checks	Internal losses or separation may be missed
Aircraft installation effects neglected	Inlet losses, combustor pressure drop, non-uniform nozzle inlet flow and structural constraints are excluded	Predicted performance is optimistic and preliminary
No CFD or experimental validation included	Results rely only on quasi-1D theoretical modelling	Design should be treated as first-pass sizing only

3.4. Conversion to Sustainable Fuel

Sustainable Aviation Fuels (SAFs) are currently starting to be used in commercial turbofan aviation, most commonly as a mixture with regular kerosene based jet fuel, commonly known as Jet – A. Most SAFs are formulated and designed to have largely the same properties as their less sustainable counterparts, this is to reduce the needed changes in engine design that would be required otherwise. As both SAFs and using RAM/SCRAM jets for commercial flights are both in their infancy little work has been done on the impacts of SAFs on these engines, therefore the modelling below is an initial estimate.

As the fuel is changing the impacts on the aircraft are somewhat isolated to the combustion chamber, in particular the mass flow rate of fuel. The properties of the fuel mix chosen and their sources are given in Table 12. Calculations for R_{mix} and γ are given in Appendix 5.

Table 12: SAF Fuel Properties

Quantity	Value	Source
H_v	42.8	[14]
R_{mix}	273	[15]
γ	1.3	[15]
C_p	1183	Calculated

Using Equation 65 below the mass flow rate of the fuel could then be calculated using the same formula and conditions as in Section 3.3.

$$\dot{m}_{fuel} = \frac{\dot{m}_{total} c_p (T_{02} - T_{01})}{q_{fuel} - c_p T_{01}} \quad (71)$$

This showed only a very slight increase from the value with the normal fuel of 1.40kg/s and 1.20×10^{-2} kg to 1.41kg/s and 1.21×10^{-2} kg/s for the RAM and SCRAM respectively in the mass flow rate of fuel required to maintain the same outlet and inlet conditions as for regular jet fuel. Meaning that the aircraft would only require a slight change in the size of the fuel tank to accommodate this change, without the impact on range. It is possible that further optimisations on the C-D nozzle and inlet components could be performed to further optimise the inlet and outlet conditions with the new fuel but given the minor change in fuel flow rate this is unlikely to have a major impact.

4. Comparison to Experimental Data

The trends found within the experimental data are promising for validation of the theory used throughout this report. Both Scramjet and Ramjet configurations rely heavily on slowing the flow down to perform combustion and subsequently speeding the flow back up through a converging diverging nozzle. It is notable to see how, in the experimental data, the flow reaches close to choking – but cannot break into a supersonic regime.

This difference in flow behaviours is due to the energy present in the flow – in the design, there is an extremely large quantity of energy backing the fluid as it approaches the throat of the nozzle, allowing the fluid to break into a supersonic regime after the nozzle. Conversely, in the experiment the driving factor does not have a large amount of energy behind it, particularly as the flow is being “pulled” instead of pushed. This means that although Mach 1 is approached at the throat, the fluid does not break into supersonic and thus decelerates through the diverging section of the nozzle – as expected of subsonic flow.

Using only low powered equipment, it was possible to approach Mach 1 and validate the underlying theory about converging nozzles utilised in the diffuser and CD nozzle sections but the supersonic flow seen in the design section could not be replicated.

5. Conclusion

This report has reviewed and validated existing theory, as well as proposed an effective design for producing a dual-mode Ramjet Scramjet engine. The experimental trends aligned extremely well with the theoretical model, which is extremely promising for validation of the overall design. There were several difficulties encountered with the project, primarily caused by the parameters imposed on the design that did not align with realistic engines found in literature. Despite this, reasonable designs were provided within possible limits for each stage.

6. Appendices

Appendix 1 – Contribution Table

Table 13: Summary of Contributions from each member

Name	Sections	Percentage Contribution
Mackenzie Fettes	Diffuser	20%
Fraser Forrest	Inlet Geometry, Experimental Data Analysis, Conversion to Sustainable Fuels and Conclusion	20%
Archie Nichols	Introduction and Combustion Chamber	20%
Alex Lansley	Convergent Divergent Nozzle	20%
Finn Thompson	Experimental data analysis/write up, Conversion to Sustainable and Abstract	20%

Appendix 2 - Raw Experimental Data

Table 14: Raw Experimental Data used for the analysis

Fan speed (rpm)	Inlet Mach Number	Pressures throughout nozzle (mbar)					Outlet Pressures (bar)
		1	2	3	4	5	
4000	0.022	4	4.8	7	11.8	22.1	0.004
6000	0.036	9.8	12.6	18.4	32.5	60.1	0.011
8000	0.047	20.1	25.2	35.4	63.8	122.7	0.018
10000	0.058	31.3	39.5	54.4	95.9	191.3	0.027
12000	0.066	49.2	60.5	80.7	136.9	295	0.044
14000	0.070	66.9	79.2	102.5	175.7	370.5	0.06
16000	0.072	88.3	104.3	135.2	218.7	439.1	0.082
18000	0.072	112.2	129.8	167.4	261.3	451.9	0.106
20000	0.072	137.3	156.7	203.7	313.7	452.4	0.134
22000	0.072	163.6	183.6	237.4	359.2	452.6	0.161
23500	0.072	183.7	204.9	264	386.3	453.1	0.181

Appendix 3 - CD Nozzle Sample Calculations

Atmospheric speed of sound

$$a_{\infty} \sqrt{1.4 \times 287 \times 216.65}$$

$$= 295.04 \text{ ms}^{-1}$$

Therefore, the freestream velocities are as follows.

Minimum acceleration force

$$\Delta V = 891.03 - 773.01 = 118.2 \text{ ms}^{-1}$$

and

$$F_{acc} = 64500 \frac{118.02}{1800} = 4228.94 \text{ N}$$

$$= 4.23 \text{ kN}$$

Scramjet cruise:

$$V_{\infty,s} = M_{\infty,s} a_{\infty}$$

$$= 3.02 \times 295.04$$

$$= 891.03 \text{ ms}^{-1}$$

Ramjet point:

$$V_{\infty,r} = M_{\infty,r} a_{\infty}$$

$$= 2.62 \times 295.04$$

$$= 773.01 \text{ ms}^{-1}$$

Using ISA, the ambient pressures are approximately,

$$P_{\infty,s} = 5.51 \text{ kPa}$$

$$P_{\infty,r} = 18.31 \text{ kPa}$$

Scramjet exit Mach number

$$M_e = \sqrt{\frac{2}{\gamma - 1} \left[\left(\frac{P_0}{P_e} \right)^{\frac{\gamma - 1}{\gamma}} - 1 \right]}$$

$$= \sqrt{\frac{2}{0.3} \left[\left(\frac{300000}{5510} \right)^{\frac{0.3}{1.3}} - 1 \right]}$$

$$\therefore M_{e,s} = 3.18$$

Nozzle area ratio

$$\frac{A_e}{A^*} = \frac{1}{M_e} \left[\frac{2}{\gamma + 1} \left(1 + \frac{\gamma - 1}{2} M_e^2 \right) \right]^{\frac{\gamma + 1}{2(\gamma - 1)}}$$

$$\frac{A_e}{A^*} = \frac{1}{3.178} (2.1869)^{3.8333}$$

$$= 6.32$$

Exit temperature

The isentropic relationship is:

$$\frac{T_0}{T_e} = 1 + \frac{\gamma - 1}{2} M_e^2$$

Hence,

$$\begin{aligned} T_e &= \frac{T_0}{1 + \frac{\gamma - 1}{2} M_e^2} \\ &= \frac{1800}{1 + 0.15(3.178)^2} \\ \therefore T_e &= 715.59K \end{aligned}$$

Exit velocity

The speed of sound is calculated by:

$$\begin{aligned} a_e &= \sqrt{\gamma R T_e} \\ &= \sqrt{1.3 \times 276 \times 715.59} \\ \therefore a_e &= 506.71 \text{ ms}^{-1} \end{aligned}$$

Therefore, the exhaust velocity is as follows.

$$\begin{aligned} V_e &= M_e a_e \\ &= 3.178 \times 506.71 \\ \therefore V_e &= 1610.56 \text{ ms}^{-1} \end{aligned}$$

Required mass flow rate

Nozzle thrust is determined using the following.

$$F = \dot{m}(V_e - V_\infty) + (P_e - P_\infty)A_e$$

At the cruise design point, the ideal expansion means

$$P_e = P_\infty$$

So the pressure-thrust term is zero and

$$F = \dot{m}(V_e - V_\infty)$$

Therefore,

$$\begin{aligned} \dot{m}_{scramjet} &= \frac{F}{V_e - V_\infty} \\ &= \frac{30300}{1610.56 - 891.03} \\ \therefore \dot{m}_{scramjet} &= 42.11 \text{ kg s}^{-1} \end{aligned}$$

The same choked flow equation for the ramjet gives 57.77kg/s.

Throat area from choking

The choked mass-flow equation is:

$$\dot{m} = A^* \frac{P_0}{\sqrt{T_0}} \sqrt{\frac{\gamma}{R}} \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma+1}{2(\gamma-1)}}$$

Rearranging for the throat area,

$$A^* = \frac{42.11\sqrt{1800}}{300000 \sqrt{\frac{1.3}{276}} \left(\frac{2}{2.3} \right)^{\frac{2.3}{0.6}}}$$

$$= 0.1483 \text{ m}^2$$

Then

$$A_e = \left(\frac{A_e}{A^*} \right) A^*$$

$$= 6.32 \times 0.1483$$

$$\therefore A_e = 0.9372 \text{ m}^2$$

Circular diameters

Assuming a circular throat and exit:

$$d_t = \sqrt{\frac{4A^*}{\pi}}$$

$$= \sqrt{\frac{4(0.1483)}{\pi}}$$

$$\therefore d_t = 0.4345 \text{ m}$$

$$d_e = \sqrt{\frac{4A_e}{\pi}}$$

$$= \sqrt{\frac{4(0.9372)}{\pi}}$$

$$\therefore d_e = 1.0924 \text{ m}$$

Appendix 4 - Diffuser

Table 15: Combustion chamber inlets from the Diffuser

Diffuser Outlet - Combustion chamber inlets					Units
Variable	symbol	Equation	Ram	Scram	-
Diffuser outlet Mach	M2	Given	0.26	1.2	-
Stagnation temperature	T_02	Combustion chamber	505.78	620.94	K
static temperature	T_2	Combustion chamber	500.70	510.64	K
stagnation pressure	P_02	Combustion chamber	461187.02	303234.69	Pa
static pressure	P_2	Combustion chamber	441458.90	129937.17	Pa

Table 16: Diffuser Outlets

Diffuser outlet					Units
Variable	symbol	Equation	Ram	Scram	-
speed of sound at outlet	a ₂	$\text{SQRT}(\gamma RT_2)$	423.85	428.04	m/s
flow velocity at outlet	u ₂	$M_2 a_2$	110.20	513.65	m/s
density at outlet	rho ₂	$P_2/(RT_2)$	3.19	0.92	kg/m ³
Area-Mach at M ₂	A ₂ /A*	$(1/M_2)^*[(2/(\gamma+1))^*(1+(\gamma-1)/2*M_2^2)]^{((\gamma+1)/(2(\gamma-1)))}$	2.34	1.03	-
Check T ₀₁	T _{01,check}	$T_2*[1+(\gamma-1)/2*M_2^2]$	503.75	576.82	K
Check P ₀₁	P _{01,check}	$P_2*[1+(\gamma-1)/2*M_2^2]^{\gamma/(\gamma-1)}$	461187.02	303234.69	Pa

Table 17: Diffuser Inlets

Diffuser Inlet					Units
Variable	symbol	Equation	Ram	Scram	-
diffuser inlet Mach number	M ₁	given	0.76	1.52	-
stagnation temperature	T _{0,1}	T ₁ =T ₂ (adiabatic)	505.78	620.94	K
static temperature	T ₁	$T_{0,1}/[1+(\gamma-1)/2*M_1^2]$	465.45	461.13	K
stagnation pressure	P _{0,1}	P _{0,1} =P _{0,1} (isentropic)	461187.02	303234.69	Pa
static pressure	P ₁	$P_{0,in}/[1+(\gamma+1)/2*M_1^2]^{\gamma/(\gamma-1)}$	321740.52	83521.98	Pa
speed of sound	a ₁	$\text{SQRT}(\gamma RT_1)$	408.66	406.76	m/s
flow velocity	u ₁	$M_1 a_1$	310.58	618.27	m/s
density	rho ₁	$P_1/(RT_1)$	2.50	0.66	kg/m ³
Area-Mach	A ₁ /A*	$(1/M_1)^*[(2/(\gamma+1))^*(1+(\gamma-1)/2*M_1^2)]^{((\gamma+1)/(2(\gamma-1)))}$	1.06	1.20	-

Table 18: Diffuser Ratios

Ratios across diffuser					Units
Variable	symbol	Equation	Ram	Scram	-
Static temperature	T_2/T_1	T_2/T_1	1.08	1.11	-
static pressure	P_2/P_1	P_2/P_1	1.37	1.56	-
stagnation pressure	$P_{01}/P_{0,1}$	P_{02}/P_{01}	1	1	-
density	ρ_2/ρ_1	ρ_2/ρ_1	1.28	1.40	-
Velocity	u_2/u_1	u_2/u_1	0.35	0.83	-
temperature rise across diffuser	dT	T_2-T_1	35.25	49.51	-
pressure rise across diffuser	dP	P_2-P_1	119718.38	46415.18	-

Scramjet (M1=1.52):

$$T_{1,s} = \frac{1769.5}{\left(1 + \frac{1.3-1}{2} * 1.52^2\right)} = 1314.12 \text{ K}$$

$$P_{1,s} = \frac{626838}{\left(1 + \frac{1.3-1}{2} * 1.52^2\right)^{\frac{1.3}{1.3-1}}} = 172654.25 \text{ Pa}$$

Ramjet (M1=0.76):

$$T_{1,r} = \frac{469.32}{\left(1 + \frac{1.3-1}{2} * 0.76^2\right)} = 431.9 \text{ K}$$

$$P_{1,r} = \frac{464015.7}{\left(1 + \frac{1.3-1}{2} * 0.76^2\right)^{\frac{1.3}{1.3-1}}} = 323713.9 \text{ Pa}$$

Scramjet inlet:

$$a_{in,s} = \sqrt{1.3 * 276 * 1314.12} = 686.66 \text{ m/s}$$

$$V_{in,s} = 1.52 * 686.66 = 1043.73 \text{ m/s}$$

$$\rho_{in,s} = \frac{172654.25}{276 * 1314.12} = 0.476 \text{ kg/m}^3$$

Ramjet Inlet:

$$a_{in,r} = \sqrt{1.3 * 276 * 431.9} = 393.66 \text{ m/s}$$

$$V_{in,s} = 0.76 * 393.66 = 299.18 \text{ m/s}$$

$$\rho_{in,s} = \frac{323713.9}{276 * 431.9} = 2.716 \text{ kg/m}^3$$

Scramjet outlet:

$$a_{2,s} = \sqrt{1.3 * 276 * 1455.21} = 722.59 \text{ m/s}$$

$$u_{2,s} = 1.2 * 722.59 = 867.1 \text{ m/s}$$

$$\rho_{in,s} = \frac{268602.3}{276 * 1455.21} = 0.669 \text{ kg/m}^3$$

Ramjet Outlet:

$$a_{2,r} = \sqrt{1.3 * 276 * 464.61} = 408.29 \text{ m/s}$$

$$V_{2,rs} = 0.26 * 408.29 = 106.16 \text{ m/s}$$

$$\rho_{in,r} = \frac{441458.9}{276 * 500.7} = 3.19 \text{ kg/m}^3$$

Appendix 5 - SAF Calculations

R_{mix}

Assuming a 14.7:1 fuel ratio as is typical for 50% SAF fuel mix and a molecular weight of SAF mixture of 0.176kg/mol [15] and of air of 0.02897kg/mol;

$$R_{mix} = m_{fuel} + \frac{m_{air} + m_{fuel}}{\frac{m_{fuel}}{MW_{fuel}} + \frac{m_{air}}{MW_{air}}} = 0.0306 \text{ kg/mol}$$

$$R_{mix} = \frac{R}{MW_{mix}} = 272 \text{ Jkg}^{-1} \text{ K}^{-1}$$

γ

Assuming a Cp of 1.2425kJ/KgK

$$\gamma_{mix} = \frac{c_p}{c_p - R_{mix}} = 1.3$$

7. References

- [1] United States Congress,, 2021 Session. (2021). *Title 14 Section § 91.817, Aeronautics and Space*. [Online]. Available: <https://www.govinfo.gov/app/details/CFR-2025-title14-vol2/CFR-2025-title14-vol2-sec91-817/summary>.
- [2] 119th Congress (2025-2026) Session. (2025, May 14th). *H.R.3410 Supersonic Aviation Modernization Act*. [Online]. Available: <https://www.govinfo.gov/app/details/BILLS-119hr3410eh>.
- [3] Boom Supersonic. "Boom - Overture." Boom Supersonic,. <http://boomsupersonic.com/overture> (accessed 23rd of April, 2026).
- [4] Spike Aerospace. "Solving the Sonic Boom Problem with Quiet Supersonic." Spike Aerospace,. <https://www.spikeaerospace.com/solving-the-sonic-boom-problem-with-quiet-supersonic-flight/> (accessed 23rd of April, 2026).
- [5] N. Kumar and V. A. Yadav, "Preliminary Design of a Ramjet Engine: An Analytical Approach," *Journal of Thermal and Fluid Science*, vol. 3, no. 2, pp. 38–49, October 2022, doi: <https://doi.org/10.26706/jtfs.3.2.20220901>.
- [6] Heritage Concorde. "Concorde Olympus 593." <https://www.heritageconcorde.com/concorde-olympus-593-mk610-engines> (accessed 23rd of April, 2026).
- [7] Heriot-Watt University. "B50EL and B51EL Compressible Flow Demonstration Laboratory Manual." Heriot-Watt University. <https://canvas.hw.ac.uk/courses/30790> (accessed 20th of April, 2026).
- [8] Heriot-Watt University. "Topic 1: Concepts and Terminology." Heriot-Watt University. <https://canvas.hw.ac.uk/courses/30790> (accessed 20th of April, 2026).
- [9] M. Nazarinia, M. Crichton, and A. Sipi. "Compressible Flow of Dry Air Tables." Heriot-Watt University. <https://canvas.hw.ac.uk/courses/30790> (accessed 20th of April, 2026).
- [10] J. T. Anderson and Lockheed Martin Skunk Works, "How Supersonic Inlets Work - Details of the SR-71 Mixed Compression Inlet Geometry and Operation," presented at the 11th AEHS Convention, San Diego, CA, 2014. [Online]. Available: <https://www.enginehistory.org/Convention/2014/SR-71Inlts/SR-71Inlts.shtml>.
- [11] Nasa Glenn Research Centre. "Earth Atmosphere Model." Nasa. <https://www.grc.nasa.gov/WWW/K-12/airplane/atmosmet.html> (accessed 23rd of April, 2026).
- [12] N. Hall. "Dynamic Pressure." NASA. <https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/dynamic-pressure-2/> (accessed 23rd of April, 2026).
- [13] C. J. Brasefield, "WINDS AND TEMPERATURES IN THE LOWER STRATOSPHERE," (in English), *Journal of Atmospheric Sciences*, vol. 7, no. 1, pp. 66–69, 01 Feb. 1950 1950, doi: [https://doi.org/10.1175/1520-0469\(1950\)007<0066:WATITL>2.0.CO;2](https://doi.org/10.1175/1520-0469(1950)007<0066:WATITL>2.0.CO;2).
- [14] *Standard Specification for Aviation Turbine Fuel Containing Synthesized Hydrocarbons*, ASTM D7566-25a, ASTM, 2025. [Online]. Available: <https://doi.org/10.1520/D7566-25A>
- [15] R. C. Boehm, L. C. Scholla, and J. S. Heyne, "Sustainable alternative fuel effects on energy consumption of jet engines," *Fuel*, vol. 304, p. 121378, 2021/11/15/ 2021, doi: <https://doi.org/10.1016/j.fuel.2021.121378>.